

Cambridge IGCSE Economics 0455

全册六段式双语系统教材 | Six-part Bilingual System Coursebook

适用考纲 | Syllabus: 2027–2029 版本 | Edition: 2026.09 全新重制版

本书按最新官方六大单元编排。每单元固定采用“单元导读 → 知识点精讲 → 核心词汇表 → 单元练习 → 练习答案与解析 → 自学检查清单”。题目依据 Cambridge 题型重新创作，不复制历年真题原文。

考试结构与本书使用方法 | Assessment and how to use this book

Paper 1 Multiple Choice: 1 hour, 40 marks, 40 questions, 30%.

Paper 2 Structured Questions: 2 hours, 80 marks, 70%. Section A contains one compulsory data-response question; Section B requires three questions from four.

全课程能力权重约为 AO1 43% / AO2 47% / AO3 10%。因此本书既讲准确知识，也反复训练 application、analysis 与有条件评价。

建议先连续阅读第 2 部分并自己重画图，再遮住第 5 部分完成单元练习，最后用第 6 部分逐项回查。

全册目录 | Contents

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单元 1 | 基本经济问题 | The basic economic problem

官方范围 | Official scope: 1.1–1.4

1. 单元导读 | Unit introduction

本单元回答经济学最先提出的问题：为什么任何个人、企业和政府都必须选择。学习顺序是“有限资源与无限欲望 → 稀缺 → 选择 → 机会成本 → PPC 模型”。PPC

把抽象取舍变成可计算、可作图的关系，也是后续专业化、经济增长和资源配置的基础。

与人教版衔接及口径差异 | Connection with Chinese curricula.

与人教版思想政治和初中生活经验相衔接：学生已经接触预算约束、劳动和生产资料；Cambridge 更强调 precise definition、next best alternative forgone、四类生产要素及 PPC

的图形推演。中文中的“成本”常指货币支出，Cambridge 的 opportunity cost 指被放弃的最佳替代收益，两者不能混用。

学习路线 | Learning route.

1. 1.1 经济问题的本质：稀缺、选择、经济物品与免费物品 – The nature of the economic problem

2. 1.2 生产要素：数量、质量、流动性与报酬 – Factors of production

3. 1.4 生产可能性曲线：效率、机会成本与经济增长 – Production possibility curves

4. 1.3–1.4 机会成本与 PPC 综合判断 – Opportunity cost and PPC applications

2. 知识点精讲 | Detailed teaching

1.1 | 经济问题的本质：稀缺、选择、经济物品与免费物品 | The nature of the economic problem

稀缺 (scarcity) 不是“东西很少”，而是资源不足以同时满足所有欲望。空气通常不稀缺，但在潜水瓶、医院氧气或严重污染情境中会变成稀缺资源。稀缺迫使家庭、企业、政府作出选择 (choice)。每一次选择的真正代价是机会成本 (opportunity cost)：被放弃的“最佳替代方案”的价值，而不是所有放弃物的总和。

例子：周六下午你可做兼职赚 120 元，或复习经济学。若选择复习，机会成本是 120 元，而非“120 元加上休息时间”。答题时写：The opportunity cost of studying is the income forgone from the best alternative use of time.

稀缺必须同时包含两个条件：人的欲望很多，而可用于满足欲望的资源有限。因此，价格很高不等于稀缺，免费物品也不一定永远不稀缺。学校只有一间实验室，政府只有一笔预算，家庭一天只有 24

小时，三者都面对资源约束。选择发生后才出现机会成本，而且机会成本只计算 next best alternative forgone (被放弃的最佳替代方案)。

假设市政府可用 2

亿元修建医院或体育馆。若医院是最佳替代用途，选择体育馆的机会成本就是医院带来的医疗服务，而不是“2 亿元”本身，因为这笔钱在两种方案中都会使用。选择还可能改变不同群体的福利：体育馆使用者受益，等待医疗的人可能受损。| GCSE 题目中的高分答案不仅写定义，还要把有限资源、互斥用途和最佳替代方案连接起来。

家庭面对 income 与 time，企业面对 labour、capital 和 finance，政府面对 tax revenue

与公共资源。三者都必须从互不相容的用途之间选择；机会成本必须从该主体实际可行的替代方案中确定。

政府税收有限 → 同一批建筑工人和材料不能同时建设医院与机场 → 选择机场 → 医院服务成为 opportunity cost → 不同居民群体受到不同影响。

Scarcity creates a constraint, choice is the decision made under that constraint, and opportunity cost measures the value of the next best option forgone. The alternative must be feasible and mutually exclusive with the chosen option. A complete answer identifies both the decision-maker and the specific benefit sacrificed.

不要把 shortage 与 scarcity 混用。Shortage 是某价格下 $Q_d > Q_s$ ，价格变化可消除；scarcity 是资源相对于欲望有限，是所有经济体长期面对的基本问题。

1.2 | 生产要素：数量、质量、流动性与报酬 | Factors of production

Land (土地)：自然资源，如耕地、矿产、森林；报酬为 rent (地租)。

Labour (劳动)：人提供的体力与脑力；报酬为 wages (工资)。

Capital (资本)：用于生产的人造物品，如机器、厂房、软件；报酬为 interest (利息)。它不是“有钱”本身。

Enterprise (企业家才能)：组织资源、承担风险、创新和作决策；报酬为 profit (利润)。

常见误区：CEO 的工作属于 labour；一辆消费者自用汽车不是 capital，但出租车公司用于提供运输服务的汽车是 capital。

知识点 01/02 | 四种生产要素与报酬：四种要素不是四种“东西” | Classifying factors by their productive function

Land 包括自然提供的资源，不只是一块土地；海洋、太阳能、矿藏都属于 land。Labour

是人的体力和脑力投入，数量之外还要看技能、健康和教育。Capital

是为生产其他产品而制造的生产资料，银行账户里的钱是 finance，不自动等于 capital。Enterprise 负责组织其他要素、承担不确定性并作出创新决策。同一个物品的分类取决于用途：家庭自用电脑是消费品，设计公司用于工作的电脑是 capital good。

知识点 02/02 | 四种生产要素与报酬：数量、质量、流动性与报酬 | Quantity, quality, mobility and factor rewards

生产能力不仅由要素数量决定，还取决于质量和 mobility。培训提升 labour quality；新机器提高 capital quality；灌溉可提高 land productivity。工人可能因住房、资格或语言限制而不能从衰退行业转到扩张行业，这叫 occupational/geographical immobility。要素报酬通常写作 land-rent、labour-wages、capital-interest、enterprise-profit，但现实中企业家也可能领取工资，考试应按其所承担的经济功能分类。

Land 看自然资源，labour 看人的体力与脑力，capital 看人为制造且用于生产的工具，enterprise 看组织与承担风险。分类取决于资源在生产中的功能，不取决于物体名称。

五名工人每天生产 200 件，labour productivity = $200/5 = 40$ 件/人。培训后四名工人生产 200 件，生产率 = 50 件/人，提高 25%。总产量没变，但单位劳动投入效率提高。

The classification of a resource depends on how it is used. A van owned by a delivery firm is capital, while the driver provides labour. Productivity is output per unit of input. An increase in the number of workers can raise total output without raising labour productivity.

Money used to buy a machine is finance; the machine itself is capital. Shares and banknotes are financial assets, not automatically factors of production.

增加工人数量提高 labour quantity；培训提高 labour quality；同样的工人生产更多才是 productivity 提高。答题时分别说明数量、质量和每单位投入产出，不能用“劳动力增加”同时代替三者。

1.4 | 生产可能性曲线：效率、机会成本与经济增长 | Production possibility curves

PPC 显示在既定资源与技术下，一个经济体能够生产两种产品的最大组合。曲线上任一点是 productively efficient (生产有效率)；曲线内表示资源闲置或失业；曲线外短期不可达到。曲线向外移动来自更多/更好的资源、教育培训、技术进步和投资；战争、自然灾害或劳动力流失会使曲线向内移动。

若 PPC 向原点凹，表示生产一种产品越多，转移资源的适配度越差，opportunity cost increases（机会成本递增）。画图题务必写坐标、两种产品、曲线与移动箭头；不要把“沿 PPC 移动”（产品组合改变）误画成“PPC 整条移动”（生产能力改变）。

知识点 01/06 | PPC 图像的读法：从坐标到结论

先画两条互相垂直的轴：横轴写 Good X（例如 computers），纵轴写 Good Y（例如 food），并标出原点 O。曲线连接两个截距：若所有资源生产食物，位于纵轴截距；若所有资源生产电脑，位于横轴截距。曲线上的 A、B、C 是不同的可行组合，不是三个“效率等级”。从 A 沿曲线向 B 移动，表示资源从食物部门转向电脑部门；生产能力没有改变，只是组合改变。

图中曲线以内的点 D 代表 productive inefficiency（生产无效率）：在不减少另一种产品的情况下仍可增加产出，原因可能是失业、机器闲置或管理低效。曲线以外的点 E 在当前资源和技术下不可达。若培训提高劳动生产率，整条 PPC 向外移；若地震摧毁厂房，整条 PPC 向内移。答题时用箭头标示 shift，并明确写出“productive potential changes”。

知识点 02/06 | 为什么曲线通常向原点凹？机会成本的推演

假设经济体只有土地、劳动和资本，生产电脑和食物。最初从食物转出少量资源时，可以先转移最适合电脑生产的工人和设备，所以电脑增加 1 单位只牺牲 1 单位食物。继续转移时，电脑部门需要使用原本更适合种植的土地和农业工人，牺牲的食物增加到 2、3、4 单位。这就是 increasing opportunity cost（机会成本递增），图上表现为越接近横轴，曲线越陡。

组合	Computers	Food	从上一个组合增加电脑	机会成本（少生产的 Food）
A	0	100	–	–
B	10	96	+10	4
C	20	88	+10	8
D	30	74	+10	14
E	40	50	+10	24

由表可见，每增加相同的 10 台电脑，放弃的食物越来越多。若题目要求“calculate opportunity cost”，要写 $\text{opportunity cost} = \text{output forgone} / \text{output gained}$ ，并说明是哪个区间；不能只写“机会成本为 14”，因为不同区间的数值不同。

知识点 03/06 | 课堂例题：区分 movement 与 shift

某国从组合 A（20 computers, 88 tonnes of food）变为组合 B（30 computers, 74 tonnes of food）。第一问：发生了什么？答案是沿 PPC 移动，电脑增加 10，食物减少 14；机会成本为 14 吨食物。第二问：若新工厂和自动化使每一种产品的最大产量都提高，如何作图？答案是 PPC outward shift；原曲线上的 A、B 仍是旧生产能力下的组合，新曲线应位于右上方。

高分英语表达：The economy moves from A to B along the same PPC, so resources are reallocated and the opportunity cost of 10 additional computers is 14 tonnes of food. In contrast, an outward shift would indicate an increase in productive potential caused by better resources or technology.

知识点 04/06 | 画图与评价的评分清单

1. 坐标轴有名称和方向；2. 曲线标为 PPC/PPC；3. 点标 A、B、inside、outside；4. 移动用沿曲线箭头，能力变化用整条曲线箭头；5. 解释原因，而非只写“向外”；6. 写出短期可行性和长期增长的区别。若资源只适合其中一种产品，PPC 可接近直线；若资源完全不能互换，机会成本近似不变。

知识点 05/06 | 生产可能性曲线：PPC 是约束图，不是需求供给图 | The PPC as a production constraint

PPC 的每一个点都对应两种产品的产量组合，并假定资源数量、资源质量和技术暂时不变。曲线上点表示所有现有资源已被充分有效利用；曲线内点表示实际产量低于潜在产量，原因可能是 cyclical unemployment、机器闲置或组织低效；曲线外点不是“永远不可能”，而是以当前生产能力不能持续达到。贸易可以让消费点位于本国 PPC 外，但并不等于本国生产点在曲线外。

知识点 06/06 | 生产可能性曲线：机会成本、曲率与偏向性移动 | Opportunity cost, curvature and biased shifts

从 A(0 computers,100 food) 到 B(10,96)，10 台电脑的机会成本是 4 单位 food；再到 C(20,88)，新增 10 台电脑的机会成本变为 8。机会成本递增的原因不是曲线“天生弯”，而是生产要素具有 specialised uses。若只有电脑技术进步，PPC

主要绕食物轴截距向外旋转；若教育普遍提高生产率，两端都可能外移。沿曲线移动表示 resource reallocation，不表示 growth。

坐标轴是两种产品，边界由现有资源、技术和生产率决定。点的位置回答 actual output，曲线位置回答 productive potential，曲线斜率所表示的取舍回答 opportunity cost。

更多资本品生产 → 当前消费品机会成本上升 → capital stock 随时间增加 → future productivity rises → PPC later shifts outwards → future consumption possibilities expand。

组合 A 为 0 台机器和 100 吨粮食，B 为 10 和 94，C 为 20 和 84。A 到 B 的 10 台机器成本 6 吨粮，B 到 C 成本 10 吨，显示 increasing opportunity cost。

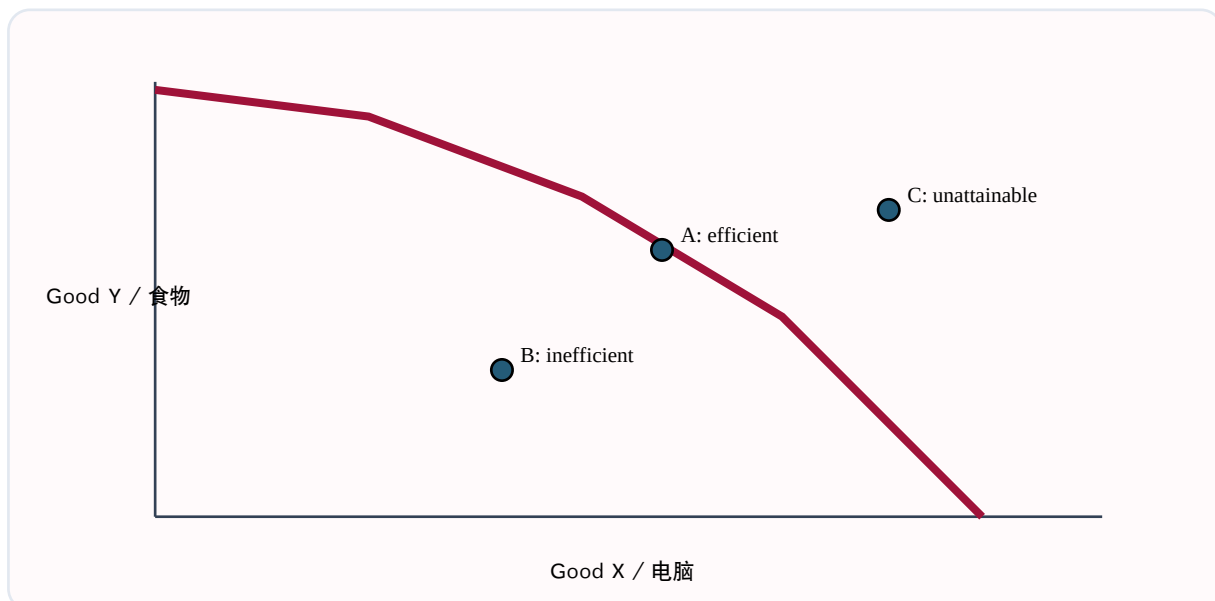
先读两条坐标轴代表的产品，再判断点在曲线上、曲线内还是曲线外。沿同一条 PPC

移动表示产品组合改变；资源数量、资源质量或技术改变才使整条 PPC 移动。图形结论必须同时写出 gained output 与 forgone output。

A bowed-out PPC reflects specialised resources. As production of one good expands, resources increasingly unsuitable for that activity are transferred, so more of the other good must be sacrificed. A straight line indicates constant opportunity cost.

曲线上的所有点都是 productively efficient，但只有符合社会偏好的组合才是 allocatively efficient。两者不能互换。

PPC 外移不保证生活水平同比提高。人口增长、污染、分配不均以及新增能力是否实际使用都会改变最终结果。



图解：PPC 上的点表示最大可行组合；曲线内点表示闲置或低效；曲线外点在当前资源与技术下不可达。沿曲线移动是组合变化，整条曲线移动才是生产能力变化。

1.3-1.4 | 机会成本与 PPC 综合判断 | Opportunity cost and PPC applications

Step 1 – 先识别位置。点在曲线内：不是“没有稀缺”，而是既定资源没有充分利用；可用 unemployment、inefficient organisation 或 recession 解释。点在曲线上：在当前技术下生产有效率，但不必然符合社会偏好。点在曲线外：当前不可行，除非贸易、技术或资源改变。

Step 2 – 再识别变化类型。“More workers receive training”提高 labour productivity, PPC outward shift; “workers move from food to computers”是 movement along PPC; “an earthquake destroys factories”是 inward shift。答题句式：An improvement in human capital raises the productive potential of the economy, so the PPC shifts outwards.

Step 3 – 最后说代价。若由 A 转向 B，机会成本是少生产的 A。请避免写“B 的机会成本是 B”；机会成本必须是放弃的另一个选项。

知识点 01/02 | 逐步判断：一个 PPC 题到底在问什么：先识别题目在改变什么 | Identifying exactly what the question changes

PPC 题只有三类核心变化：点的位置、沿曲线移动、整条曲线移动。失业下降使生产点由曲线内向曲线靠近；军用品增加且消费品减少是沿同一曲线移动；技术、劳动力、资本存量或自然资源改变才移动整条曲线。题干出现“产量上升”并不自动说明 PPC 外移，因为利用闲置资源也能提高实际产量。

知识点 02/02 | 逐步判断：一个 PPC 题到底在问什么：用反事实检查答案 | Checking a PPC conclusion with a counterfactual

问自己：资源和技术是否改变？若没有，就不能画新

PPC。再问：两种产品是否一增一减？若是，多半是沿曲线移动。若两种最大产量都提高，才通常整体外移。最后区分 short-run actual output 与 long-run productive potential。

题干出现 reallocate resources 时分析沿曲线移动；出现 unemployment falls 时分析曲线内点向边界移动；出现 technology、education、migration 或 disaster 时才判断 PPC 是否外移、内移或偏向性旋转。

旧 PPC 最大生产 50 台机器或 100 吨粮食；技术只改善机器生产，使最大机器升至 80、粮食仍为 100。图形绕粮食截距向外旋转，而不是平行移动。

先读两条坐标轴代表的产品，再判断点在曲线上、曲线内还是曲线外。沿同一条 PPC

移动表示产品组合改变；资源数量、资源质量或技术改变才使整条 PPC 移动。图形结论必须同时写出 gained output 与 forgone output。

“Output rises”不必然等于“economic growth”。使用闲置资源提高 actual output 与 productive potential 增长必须分开。

2027–2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 经济物品与免费物品 | Economic and free goods

经济物品 (economic good) 使用稀缺资源生产，数量有限，因此取得它必然涉及机会成本。免费物品 (free good) 在特定时间与地点相对充足，取得时没有机会成本，例如无人居住海岸的空气。分类取决于情境：城市中的清洁空气会因污染治理需要资源而成为 economic good。

English explanation. An economic good is scarce in relation to demand and has an opportunity cost. A free good is sufficiently abundant that consuming more does not require society to give up another output. “Free of charge” is not the same as a free good: state education may have no price at the point of use, but teachers, buildings and time are scarce.

补充知识点 02/02 | 三个基本经济问题 | What, how and for whom to produce

稀缺迫使社会决定生产什么、如何生产、为谁生产。生产什么涉及消费品与资本品的取舍；如何生产涉及 labour-intensive 与 capital-intensive 方法；为谁生产涉及收入、购买力和政府分配。三问不是背诵口号，而是同一资源约束在产出结构、生产方法和分配结果上的三个表现。

English explanation. Scarcity creates three allocation questions. What to produce concerns the composition of output. How to produce concerns the combination of factors and technology. For whom to produce concerns who receives the goods and services. Different economic systems answer these questions in different ways.

3. 核心词汇表 | Core vocabulary (共 26 词)

English term	中文准确口径	Exam-ready meaning
scarcity	稀缺	A condition in which limited resources cannot satisfy unlimited wants.
choice	选择	A decision between alternative uses of scarce resources.
opportunity cost	机会成本	The next best alternative forgone when a choice is made.
economic good	经济品	A product that is scarce and therefore has an opportunity cost.
free good	自由物品	A product available without using scarce resources in the relevant situation.
wants	欲望	Human desires for goods and services.
land	土地 / 自然资源	Natural resources used in production.
labour	劳动	Human physical and mental effort used in production.
capital	资本品	Man-made resources used to produce other goods and services.
enterprise	企业家才能	The ability to organise resources and take business risks.
productivity	生产率	Output produced per unit of input.
mobility of labour	劳动力流动性	The ability of workers to change occupation or location.
investment	投资	Spending on capital goods that increases productive capacity.
production possibility curve (PPC)	生产可能性曲线	The maximum attainable combinations of two products with given resources and technology.
productive efficiency	生产效率	Producing the maximum output possible from available resources.
allocative efficiency	配置效率	Producing the combination of goods and services most wanted by society.
unattainable	当前不可达到的	Outside present productive capacity.
economic growth	经济增长	An increase in an economy's productive capacity or real output.
resource reallocation	资源重新配置	Moving factors of production from one use to another.
increasing opportunity cost	机会成本递增	Giving up progressively more of one product to obtain extra units of another.
movement along a PPC	沿PPC移动	A change in the product mix with productive capacity unchanged.
shift of a PPC	PPC整体移动	A change in maximum productive capacity.
unemployment	失业	People willing and able to work who cannot find employment.
spare capacity	闲置生产能力	Resources that are available but not currently used.

English term	中文准确口径	Exam-ready meaning
capital goods	资本品	Goods used to produce other goods and services.
consumer goods	消费品	Goods that directly satisfy consumers' wants.

4. 单元练习 | Unit practice (共 16 题)

练习 01/16

Question. A council can build a library, repair a road or leave the money in a bank. It builds the library; road repair was its second preference. What is the opportunity cost?

练习 02/16

Question. Explain why economic growth does not remove scarcity.

练习 03/16

Question. Which change definitely improves the quality of labour: more mineral deposits, worker training, a larger factory, or a lower interest rate?

练习 04/16

Question. Analyse how an increase in the quantity and quality of capital may affect an economy.

练习 05/16

Question. An economy moves from a point inside its PPC to a point on the curve. What is the most likely explanation?

练习 06/16

Question. Explain how investment in capital goods can affect current and future living standards.

练习 07/16

Question. Draw a PPC and show (i) unemployed resources becoming fully used and (ii) an improvement in technology affecting both products. Explain why the two changes are different. [6]

练习 08/16

Question. What happens to the PPC when unemployed workers find jobs but the labour force and productivity are unchanged?

练习 09/16

Question. Analyse the PPC effects of a government transferring resources from consumer goods to capital goods.

练习 10/16

Question. An economy moves from a point inside its PPC to a point on the curve. Does this prove that its resources have increased? Explain. [3]

练习 11/16

Question. Which statement best distinguishes a free good from a good supplied free of charge?

A A free good has no market price. B A free good has no opportunity cost. C A free good is provided by government. D A free good is wanted by nobody.

练习 12/16

Question. A government uses land to build a hospital instead of the next-preferred school. What is the opportunity cost?

A the wages of hospital workers B the price of the land C the benefit of the school forgone D all other projects considered

练习 13/16

Question. An economy moves from A (0 machines, 100 tonnes of food) to B (10 machines, 94 tonnes) and then C (20 machines, 84 tonnes). Calculate the opportunity cost of the first and second additional 10 machines. What does the change show? [4]

练习 14/16

Question. Explain why a point inside a PPC does not mean that scarcity has disappeared. [4]

练习 15/16

Question. Explain two ways in which education can change an economy's PPC. [6]

练习 16/16

Question. Discuss whether an outward shift of a country's PPC will always raise living standards. [8]

5. 练习答案与解析 | Answers and explanations (共 16 题)

答案 01/16

正确答案 | Correct answer. The benefit from repairing the road. The bank deposit is another alternative but is not the next best one.

答案 02/16

参考答案 | Model answer. Growth shifts productive capacity outwards and allows more wants to be satisfied. However, human wants can continue to expand and resources remain limited relative to those wants. The economy therefore still has to choose how to use labour, land and capital, so opportunity cost remains.

答案 03/16

正确答案 | Correct answer. Worker training, because it increases human capital and the skills embodied in labour.

答案 04/16

参考答案 | Model answer. More and better machinery can raise output per worker. Higher productivity may lower average cost, enabling firms to reduce prices or increase profit. Competitiveness and exports may rise, increasing real GDP. The effect depends on workers having the skills to operate the capital and on sufficient demand.

答案 05/16

正确答案 | Correct answer. Unemployment or inefficiency has fallen. Existing productive capacity is used more fully; the PPC itself need not shift.

答案 06/16

参考答案 | Model answer. Allocating more resources to capital goods requires less current production of consumer goods, so present consumption may fall. The larger capital stock can raise productivity and productive potential later, shifting the PPC outwards. Future real income and consumption may then increase, provided the capital is useful and fully employed.

答案 07/16

正确答案 | Correct answer. Label both axes and draw one PPC. For (i), move an inside point to a point on the same boundary: actual output rises because idle resources are used, but productive capacity is unchanged. For (ii), draw a second PPC outside the first and label the outward shift: improved technology raises the maximum attainable combinations.

答案 08/16

正确答案 | Correct answer. The production point moves towards the existing PPC. Productive potential is unchanged.

答案 09/16

参考答案 | Model answer. In the current period, the economy moves along its PPC towards more capital goods and fewer consumer goods. The sacrificed consumer goods are the opportunity cost. If capital formation raises future productivity, the future PPC may shift outwards, allowing more of both goods.

答案 10/16

正确答案 | Correct answer. No. The movement can result from lower unemployment or better use of existing resources. Actual production rises toward the existing maximum, but the PPC itself shifts only if the quantity or quality of resources, productivity or technology changes.

答案 11/16

正确答案 | Correct answer. B – A free good has no opportunity cost. A government service may be free at the point of use but still use scarce labour, buildings and finance, so it is an economic good.

答案 12/16

正确答案 | Correct answer. C – The opportunity cost is the benefit from the next best alternative forgone, which is the school. It is not the sum of every rejected option.

答案 13/16

正确答案 | Correct answer. $A \rightarrow B$ costs $100 - 94 = 6$ tonnes of food. $B \rightarrow C$ costs $94 - 84 = 10$ tonnes. The opportunity cost rises from 6 to 10 tonnes, showing increasing opportunity cost as resources less suited to machine production are transferred.

答案 14/16

参考答案 | Model answer. A point inside the PPC shows that some resources are unemployed or used inefficiently, so actual output is below potential output. Moving towards the frontier can raise both outputs, but the frontier is still limited. Once resources are fully and efficiently used, producing more of one product requires sacrificing another, so scarcity and choice remain.

答案 15/16

参考答案 | Model answer. Education can improve labour quality and productivity, allowing more output from the same number of workers and shifting the PPC outward. It can also improve occupational mobility, enabling unemployed workers to fill expanding jobs; this may first move actual production towards the existing PPC and, if skills raise productive capacity, shift the frontier outward over time.

答案 16/16

参考答案 | Model answer. It increases productive potential, so the economy can produce more consumer goods and services, raise incomes and possibly collect more tax for public services. However, actual output rises only if the extra capacity is used. Gains may be unevenly distributed, additional production may create pollution, and a shift caused by capital-goods investment may require lower current consumption. Living standards are most likely to rise when capacity is used productively, output matches preferences and environmental and distributional costs are controlled.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能用 finite resources relative to unlimited wants 准确定义 scarcity
- ☐ 能区分 economic good、free good 与 free-of-charge service
- ☐ 能在个人、企业和政府情境中找出 next best alternative forgone
- ☐ 能分类 land、labour、capital、enterprise 并说明各自报酬
- ☐ 能解释生产要素数量或质量改变如何影响生产能力
- ☐ 能画出完整坐标、曲线和点位的 PPC
- ☐ 能解释曲线上、曲线内和曲线外点的含义
- ☐ 能从相邻组合计算机会成本并解释递增原因
- ☐ 能区分 movement along PPC、movement towards PPC 与 shift of PPC
- ☐ 能区分 productive efficiency 与 allocative efficiency

单元 2 | 资源配置 | The allocation of resources

官方范围 | Official scope: 2.1–2.10

1. 单元导读 | Unit introduction

本单元建立 IGCSE 微观经济学的核心模型。需求和供给分别解释消费者与企业的计划；价格机制把两者协调到均衡；弹性衡量反应幅度；市场失灵解释为什么自由市场结果可能无效率；政府干预则需要同时分析目标、传导与副作用。学习时必须一直区分“沿曲线移动”和“整条曲线移位”。

与人教版衔接及口径差异 | Connection with Chinese curricula. 与人教版“市场调节与宏观调控”衔接。Cambridge 要求更精确地区分 demand/quantity demanded、supply/quantity supplied，并用图回答价格和数量怎样变化。中文材料有时把“需求增加”泛指购买量变大，但在 Cambridge 图形题中 increase in demand 只表示整条需求曲线右移。

学习路线 | Learning route.

1. 2.1 市场、买方与卖方 – Markets, buyers and sellers
2. 2.2 需求、个人需求与市场需求 – Demand and market demand
3. 2.2 需求量变化与需求曲线移位 – Movements and shifts in demand
4. 2.3 供给及其决定因素 – Supply and its determinants
5. 2.4–2.5 价格机制、均衡、短缺与过剩 – The price mechanism and equilibrium
6. 2.5 需求与供给同时变化 – Simultaneous changes in demand and supply
7. 2.6–2.7 PED、PES 与销售收入 – Elasticity and revenue
8. 2.9–2.10 市场失灵与政府干预 – Market failure and government intervention
9. 2.8–2.10 市场经济、计划经济与混合经济 – Economic systems and the mixed economy

2. 知识点精讲 | Detailed teaching

2.1 | 市场、买方与卖方 | Markets, buyers and sellers

市场（market）是使买者与卖者相互接触、进行交换的一种安排；它可以是实体市场，也可以是网络平台。买方通常是 consumers，他们购买商品与服务满足需要；卖方通常是 firms，他们组织生产要素并提供产品。市场不要求买卖双方在同一地点。

English explanation. A market is an arrangement that brings buyers and sellers into contact. Buyers express demand; sellers express supply. Their decisions interact through prices. In Cambridge questions, identify the product market first and distinguish a consumer from a producer.

知识点 01/02 | 市场、买方与卖方：市场是一种安排 | A market is an arrangement, not only a place

市场不一定有店铺或固定地点。网约车平台、外汇市场和劳动力市场都让潜在买者与卖者交换。买者形成 demand，卖者形成 supply，价格承担 rationing、signalling 和 incentive 功能。分析前要先界定市场：讨论“饮料市场”与“无糖饮料市场”可能得到不同的竞争者和替代品。

知识点 02/02 | 市场、买方与卖方：市场参与者的目标与约束 | Objectives and constraints of buyers and sellers

消费者通常希望在预算内提高 satisfaction；企业希望实现利润或其他目标，但受成本、技术和竞争限制。政府也可能作为买方采购疫苗，或作为卖方提供交通服务。市场中的“买方”和“卖方”是经济角色，不是固定身份；家庭在产品市场是买方，在劳动力市场却出售 labour services。

Buyers 形成 demand, sellers 形成 supply; 市场可以是实体地点, 也可以是线上或其他使双方接触的安排。价格、数量与交易时期必须明确, 才构成可比较的市场信息。

A market includes potential transactions, not only completed sales. Its boundaries matter because a narrow definition may show few competitors, while a broad definition may include many substitutes. Online interaction is sufficient to bring buyers and sellers into contact.

2.2 | 需求、个人需求与市场需求 | Demand and market demand

需求 (demand) 是消费者在一定时期、不同价格下愿意并且能够购买的数量。个人需求是一个消费者的需求; 市场需求是在每一个价格水平把所有消费者的需求量横向相加。若价格为 \$10, A 购买 500 单位, B 购买 400 单位, 则市场需求量为 900。

English explanation. Demand is the willingness and ability to buy a product at different prices over a period of time. Market demand is the horizontal sum of all individual demand: quantities, not prices, are added at each price.

知识点 01/02 | 需求、个人需求与市场需求: Demand 包含 willing、able、price 与 time | The four elements of an accurate demand definition

想买但付不起不是有效需求, 有能力但不愿买也不是需求。需求表列出一定时期内不同价格对应的 quantity demanded, 曲线只是把表格图形化。需求定律说明 *ceteris paribus* 下价格越低, 需求量通常越大, 因为替代效应和收入效应共同作用。

知识点 02/02 | 需求、个人需求与市场需求: 市场需求为何是横向加总 | Why market demand is a horizontal sum

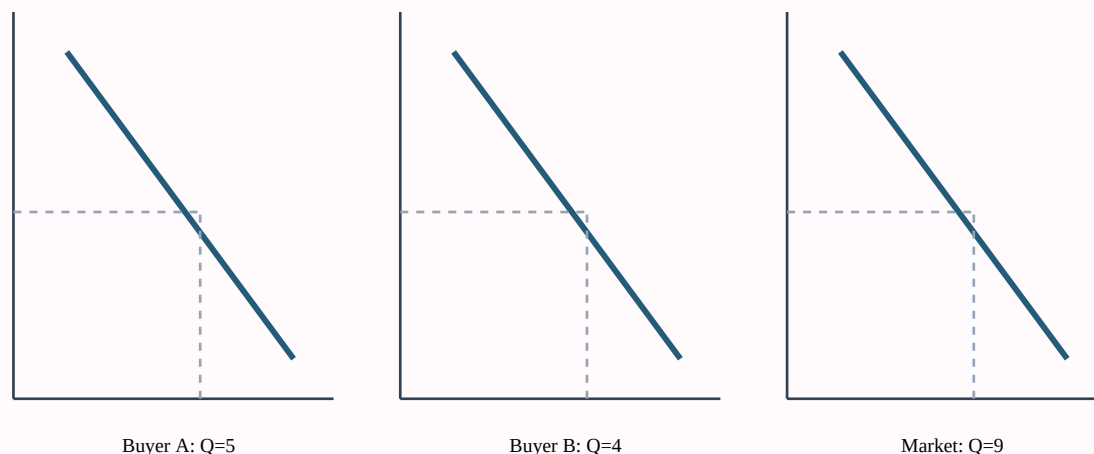
同一个价格下, 把每位消费者愿意购买的数量相加。例如 \$10 时 A 买 5 个、B 买 4 个, market quantity demanded 为 9 个; 不能把两个 \$10 相加为 \$20。若第三位消费者在该价位愿意买 2 个, 市场需求量变成 11, 整条市场需求曲线也可能向右。

Individual demand 的主体是一名消费者; market demand 的主体是同一市场内全部消费者。纵轴价格保持相同, 横轴数量逐人相加, 单位与时期也必须一致。

价格 \$12 时 A 买 3、B 买 5、C 买 0, 市场需求量 8; 价格 \$8 时分别买 5、8、2, 市场需求量 15。每行加数量, 不能加价格。

在每一个完全相同的价格上, 把所有消费者的 quantity demanded 横向相加。价格不能相加, 曲线斜率也不能直接相加; 市场曲线上的每一点都来自同价位的数量总和。

Willingness without purchasing power is a desire, not effective demand. Ability without willingness also produces no demand. The time period matters: ten units per day and ten units per year describe very different demand.



图解：市场需求是在同一价格水平横向相加每位消费者的数量；相加的是 quantities，不是 prices。

2.2 | 需求量变化与需求曲线移位 | Movements and shifts in demand

商品自身价格下降会导致 extension in demand / increase in quantity

demanded，表现为沿同一条需求曲线向右下移动；自身价格上升会导致 contraction。收入、相关商品价格、广告、偏好、人口结构、天气和未来价格预期属于非价格决定因素，它们改变整条需求曲线。正常品收入增加使需求右移；劣等品收入增加可能使需求左移。替代品价格上涨会增加本商品需求；互补品价格上涨会减少本商品需求。

English explanation. A change in the product's own price causes a movement along the existing demand curve. A non-price determinant causes the whole curve to shift. Write "quantity demanded increases" for an extension and "demand increases" for a rightward shift.

知识点 01/02 | 需求量移动与需求曲线移位：一条曲线还是两条曲线 | One demand curve or a shifted curve

判断的第一问永远是：变的是商品自己的价格，还是其他条件？Own price change 只改变 quantity demanded，点沿原曲线移动；non-price determinant 改变 demand，整条曲线移动。价格下降形成 extension in demand，价格上涨形成 contraction in demand。收入、偏好、广告、人口、预期和相关商品价格不在纵轴上，所以需要画新曲线。

知识点 02/02 | 需求量移动与需求曲线移位：相关商品与收入的方向判断 | Income and related-goods effects on demand

咖啡与茶是 substitutes：咖啡价格上升，部分消费者改买茶，茶的 demand increases。汽车与汽油是 complements：汽油价格上升使驾车总成本增加，汽车需求下降。收入上升令 normal good 需求增加，却可能令 inferior good 需求减少。先识别关系，再写消费者选择改变，最后判断 $D_0 \rightarrow D_1$ 的方向。

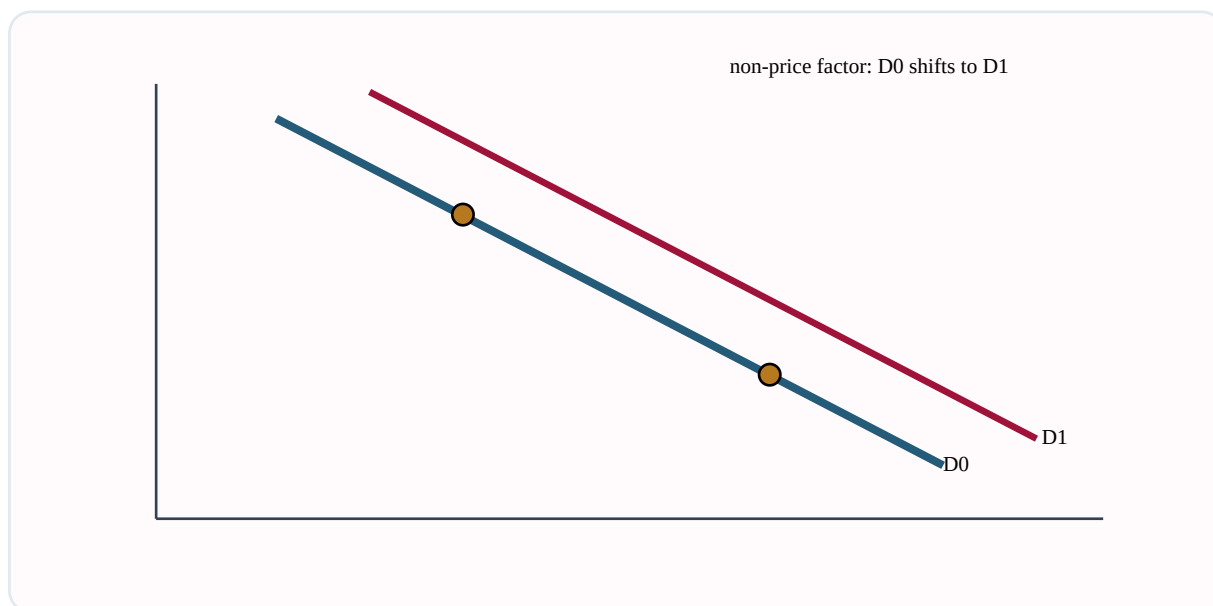
Own price 位于需求图纵轴，因此其变化表现为同一 D 上的点移动。Income、taste、population、expectations 及 related goods prices 不在坐标轴上，因此改变每一价格下的购买计划，使整条 D 移位。

tea price rises $\rightarrow$ coffee is a substitute $\rightarrow$ consumers switch from tea to coffee $\rightarrow$ coffee demand increases $\rightarrow$ coffee demand curve shifts right $\rightarrow$ coffee equilibrium price and quantity tend to rise.

商品自身价格改变时，标出同一条 D 曲线上的 A、B 两点；收入、偏好、人口、预期或相关商品价格改变时，才画 D_0 到 D_1 。作答时必须用 movement along 与 shift 的不同术语。

The curve represents the relationship between the product's own price and quantity demanded. Therefore, a change in any other determinant changes the relationship itself and requires a new curve. State both the determinant and why willingness or ability to buy changes.

“价格下降使需求增加”在 Cambridge 精确口径中应写 quantity demanded increases 或 extension in demand。



图解：商品自身价格变化造成沿 D 移动；收入、偏好、人口及相关商品价格等非价格因素使 D 整体移动。

2.3 | 供给及其决定因素 | Supply and its determinants

供给 (supply) 是在一定时期、不同价格下厂商愿意并且能够出售的数量。商品自身价格变化造成沿供给曲线移动；生产成本、技术、间接税、补贴、其他产品利润、企业数量、天气、战争和原材料发现等会使整条供给曲线移动。成本下降或技术改善使供给右移；间接税提高单位成本，使供给左移；补贴使供给右移。

English explanation. Supply is the willingness and ability of producers to sell a product at different prices over a period of time. A rise in the product's own price causes an extension in supply. Lower costs, better technology or a subsidy increase supply.

知识点 01/02 | 供给与供给决定因素：Supply 看的是厂商计划销售量 | Supply as producers planned sales

库存很多不等于市场供给一定多；供给指厂商在不同价格下愿意且能够出售的数量。价格上涨通常提高边际收益，使扩产更有吸引力，形成 extension in

supply。若生产能力、成本或政策改变，则每个价格下都可能愿意出售更多或更少，曲线才发生 shift。

知识点 02/02 | 供给与供给决定因素：决定因素的机制不能只背方向 | How costs, technology and policy shift supply

工资、能源和原材料价格下降会降低 unit cost，提高每单位利润，因此 supply increases。技术改善提高 productivity，也会降低 average cost。Indirect tax 提高单位成本，使 S 左移；subsidy 降低有效成本，使 S 右移。好天气提高农作物产量；灾害、战争或资源枯竭限制生产。其他产品更有利可图时，企业可能转移资源，当前产品供给减少。

Supply 的主体是 producers，核心是不同价格下 planned quantity supplied。Own price 改变利润边际并造成沿 S 移动；input costs、technology、tax、subsidy、weather 与 number of firms 改变整条 S。

electricity cost falls $\rightarrow$ unit production cost falls $\rightarrow$ profit per unit at every price rises $\rightarrow$ firms are willing and able to sell more $\rightarrow$ supply shifts right $\rightarrow$ equilibrium price tends to fall and quantity rises。

纵轴是 price，横轴是 quantity。先找 demand 与 supply 的交点，再从价格线分别读 Qd 和 Qs。曲线移动时先说明是哪一个非价格因素改变，再比较新旧 equilibrium price 与 equilibrium quantity。

Supply describes planned sales, not all existing stock. A warehouse may contain goods that the owner is unwilling to sell at the current price. Explain shifts through costs, profitability and productive capacity rather than memorising directions only.

“供给增加”不能仅由产品自身价格上涨造成；那是 quantity supplied 增加。

2.4-2.5 | 价格机制、均衡、短缺与过剩 | The price mechanism and equilibrium

价格机制通过价格变化协调消费者和企业的决定，并回答生产什么、如何生产和为谁生产。均衡出现在 quantity demanded equals quantity supplied。低于均衡价时出现 shortage，买方竞争产生价格上升压力；高于均衡价时出现 surplus，库存增加产生价格下降压力。

English explanation. The price mechanism allocates resources through changes in demand, supply and price. A shortage causes upward pressure on price; a surplus causes downward pressure. The market moves towards equilibrium where planned purchases equal planned sales.

知识点 01/10 | 价格机制到底是什么 | How the price mechanism coordinates a market

价格机制 (price mechanism) 是价格根据 demand 与 supply 的变化而调整，并据此配置稀缺资源的过程。它不是某个人发出命令，而是大量消费者与企业分别作出决定后，通过价格相互协调。完整分析包含三层：外部事件先改变 demand 或 supply；市场随后出现 shortage 或 surplus；价格再发生变化，使 quantity demanded 和 quantity supplied 沿原曲线调整，市场趋向新均衡。

例如某城市突然流行电动自行车，需求曲线右移。在原价格下 Q_d 大于 Q_s ，门店很快售罄。短缺向卖方传递“消费者愿意购买更多”的信息，价格可能上涨，企业也更愿意进货或扩产，最终形成较高的 equilibrium price 与较大的 equilibrium quantity。这条链比“需求增加，所以价格上涨”更完整。

知识点 02/10 | 信号功能 | Signalling function

价格像一个不断更新的信息系统。价格上涨可能表示商品变得相对稀缺，原因可能是需求增加或供给减少；价格下降可能表示需求减弱或供给增加。企业不需要知道每一位消费者为什么改变选择，也能从销量、库存和价格观察市场信号。不过，单看价格不能确定曲线移动原因：同样是 price rise，可能来自 D right，也可能来自 S left，两者对 equilibrium quantity 的影响相反。

English explanation. The signalling function means that changes in price communicate information about relative scarcity. A rising price may signal stronger demand or reduced supply. Producers should also consider whether the change is temporary, the cost of expansion and expected future demand.

知识点 03/10 | 激励功能 | Incentive function

价格上升通常提高每单位潜在 revenue 和 profit，形成增产激励。现有企业可能增加 workers' hours、使用 spare capacity 或购买 machinery；较长时期内，新企业可能 enter the market。于是 quantity supplied 先沿供给曲线增加，若企业数量或生产能力增长，整条 supply curve 还可能右移。持续低价和亏损则促使企业减产或退出，让 labour、capital 和 land 转向回报更高的用途。

激励并不保证供给立刻大幅增加。鲜花、农作物和住房需要生产时间，PES 较低；制造商若有库存和闲置设备，反应可能更快。高价也可能由间接税造成，并不表示生产者实际收入同比上涨。因此评价要考虑 time period、PES、capacity 和 costs。

English explanation. The incentive function changes behaviour. A higher market price can raise expected profit, encouraging existing firms to expand and new firms to enter. The speed and size of the response depend on price elasticity of supply, spare capacity, stocks and the production period.

知识点 04/10 | 配给功能 | Rationing function

当商品数量有限，价格上涨会减少 quantity demanded，把产品分配给 willing and able to pay 的消费者。演唱会座位、城市中心住房或干旱时期的水都可能通过价格配给。这里的 able to pay 很重要：市场配给依据购买力，不是依据“谁最需要”。所以价格机制可能有效消除 shortage，却产生 equity 问题；低收入家庭可能买不起必需品。政府可使用 subsidy、direct provision 或 maximum price，但这些政策也可能造成财政成本或新的短缺。

English explanation. The rationing function allocates a scarce product among competing consumers. A higher price reduces quantity demanded, so the available supply goes to those willing and able to pay. This may restore equilibrium, but it does not necessarily produce a fair distribution.

知识点 05/10 | 价格机制怎样回答三个基本经济问题 | How prices answer the three economic questions

生产什么 | What to produce? 消费者通过 demand

表达偏好。需求增加通常使价格与利润上升，企业把资源转入该市场；需求下降则使资源退出。准确说法是 consumer sovereignty works through demand, price and profit signals。

如何生产 | How to produce? 企业比较 labour、capital、land

等要素的相对价格和生产率，选择成本较低且符合质量要求的方法。若工资相对机器价格上升，企业可能采用 capital-intensive

production；但机器购置成本、技能、产量规模和质量也会影响决定，所以不能简单写“永远选最便宜要素”。

为谁生产 | For whom to produce? 商品主要流向愿意且能够支付市场价格的人，购买力来自

wages、rent、interest、profit、wealth 或政府转移。市场按 income and purchasing power 分配，而不是按 need 分配，这正是效率与公平之间的重要张力。

English summary. Consumer demand and profit signals influence what is produced. Relative factor prices and productivity influence how it is produced. Income and purchasing power influence for whom it is produced. These outcomes may be efficient, but they are not necessarily equitable.

知识点 06/10 | 均衡是一组价格与数量 | Equilibrium price and equilibrium quantity

均衡点 E 同时给出 equilibrium price P_e 和 equilibrium quantity Q_e 。 P_e 以下 $Q_d > Q_s$ ，出现 shortage； P_e 以上 $Q_s > Q_d$ ，出现 surplus。价格变化使需求量和供给量沿各自曲线移动，直到 planned purchases equal planned sales。均衡不是“公平价格”，也不保证每个人满意；它只表示市场没有内在的价格变化压力。

知识点 07/10 | 供过于求 | Surplus: $Q_s > Q_d$

假设 equilibrium price 是 \$6、quantity 是 100。若价格为 \$8，消费者计划购买 70，企业计划出售 130，因此 $\text{surplus} = Q_s - Q_d = 60$ 。库存积压使企业降价；价格下降造成 extension in demand 和 contraction in supply，差额逐渐缩小，直到回到均衡。D 与 S 都没有移动，因为引起调整的是商品自身价格。

English chain. Price is above equilibrium $\rightarrow$ quantity supplied exceeds quantity demanded $\rightarrow$ unsold stocks accumulate $\rightarrow$ sellers reduce price $\rightarrow$ quantity demanded extends and quantity supplied contracts $\rightarrow$ the surplus disappears at equilibrium.

口诀 $Q_s > Q_d = \text{surplus}$ 只负责快速判断；完整答案还要写 unsold stock 和 downward pressure on price。

知识点 08/10 | 供不应求 | Shortage: $Q_d > Q_s$

若价格为 \$4，消费者计划购买 140，企业只计划出售 60，因此

$\text{shortage} = Q_d - Q_s = 80$ 。消费者竞争、排队或缺货使价格承受上升压力；价格上涨造成 contraction in demand 和 extension in supply，直到 $Q_d = Q_s$ 。若 maximum price 阻止价格上升，shortage 可能持续，并出现 waiting lists、black markets 或 non-price rationing。

English chain. Price is below equilibrium $\rightarrow$ quantity demanded exceeds quantity supplied $\rightarrow$ excess demand creates upward pressure on price $\rightarrow$ quantity demanded contracts and quantity supplied extends $\rightarrow$ equilibrium is restored.

口诀 $Q_d > Q_s = \text{shortage}$ 。Shortage 不是“供给减少”的同义词，而是某一个价格下 Q_d 与 Q_s 的差额。

知识点 09/10 | 用供需数据验证两张图 | Verifying surplus and shortage with data

Price \$8: Q_d 70, Q_s 130 $\rightarrow$ surplus 60 $\rightarrow$ downward pressure on price。

Price \$6: Q_d 100, Q_s 100 $\rightarrow$ equilibrium $\rightarrow$ no tendency for price to change。

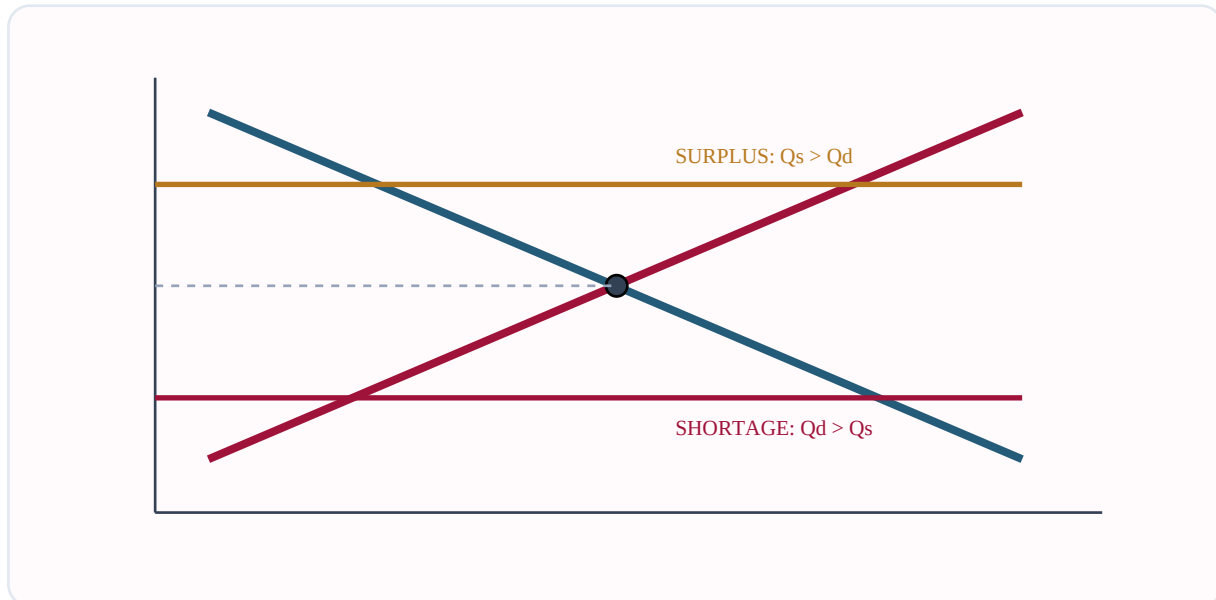
Price \$4: Q_d 140, Q_s 60 $\rightarrow$ shortage 80 $\rightarrow$ upward pressure on price。

读表顺序：在同一价格比较 Q_d 和 Q_s ；用较大者减较小者计算

gap；最后判断价格方向。不要用不同价格的两行数据相减，也不要写 shortage 写成负数。

知识点 10/10 | 图表题的得分细节 | Required labels and adjustment arrows

两张图都标 Price 纵轴、Quantity 横轴、向下的 D、向上的 S、交点 E 和均衡价 P_e 。Surplus 图在 P_e 上方画 P1：左边是 Q_d 、右边是 Q_s ，中间标 surplus。Shortage 图在 P_e 下方画 P1：左边是 Q_s 、右边是 Q_d ，中间标 shortage。只画水平线而不标 Q_d/Q_s ，阅卷人无法确认你是否理解差额。



图解：P 高于 P_e 时 $Q_s > Q_d$ ，产生 surplus 与降价压力；P 低于 P_e 时 $Q_d > Q_s$ ，产生 shortage 与涨价压力。

2.5 | 需求与供给同时变化 | Simultaneous changes in demand and supply

当需求和供给同时右移时，均衡数量一定增加，但均衡价格方向取决于两条曲线移动的相对幅度。如果需求增加幅度大于供给增加，价格上升；若供给增加幅度更大，价格下降。若一条右移、一条左移，则可能只能确定价格或数量中的一个。

English explanation. When both demand and supply increase, equilibrium quantity rises, but the effect on equilibrium price is uncertain. It depends on the relative size of the two shifts. State what is certain and what is indeterminate.

知识点 01/02 | 需求和供给同时移动：先单独判断，再合并 | Combining separate demand and supply effects

同时移动题不能凭直觉直接下结论。先只看 demand shift 对 P、Q 的作用，再只看 supply shift，最后比较方向。 $D \uparrow$ 使 $P \uparrow Q \uparrow$ ； $S \uparrow$ 使 $P \downarrow Q \uparrow$ ，合并后 Q 一定 $\uparrow$ ，P 不确定。 $D \downarrow$ 与 $S \downarrow$ 使 Q 一定 $\downarrow$ ，P 不确定。 $D \uparrow$ 与 $S \downarrow$ 使 P 一定 $\uparrow$ ，Q 不确定； $D \downarrow$ 与 $S \uparrow$ 使 P 一定 $\downarrow$ ，Q 不确定。

知识点 02/02 | 需求和供给同时移动：“不确定”也需要经济解释 | Explaining an indeterminate price or quantity

不确定不是不会做，而是缺少 relative magnitude。若 demand increase 比 supply increase 大，价格上升；若 supply increase 更大，价格下降；幅度相同则价格可能不变。图中可以画两种可能的新交点，或在文字中清楚说明条件。

当 D 与 S 同时移动，equilibrium price 和 quantity 分别受到两个方向影响。先隔离每一次 shift，再把对同一结果变量的方向合并；同向可确定，反向则取决于 relative magnitude。

healthy diet campaign $\rightarrow$ tomato demand right $\rightarrow P \uparrow Q \uparrow$ ；good harvest $\rightarrow$ tomato supply right $\rightarrow P \downarrow Q \uparrow$ ；合并后 Q $\uparrow$ ，P depends on relative shifts。

先只移动 demand 并记录价格、数量方向，再恢复原图只移动

supply，最后把两个结果合并。若两个变化对某一变量方向相反，就只能写 depends on the relative size of the shifts。

Do not add or average separate equilibrium prices. The final equilibrium is the intersection of the two new curves. State the certain variable first and then explain why the other is indeterminate.

2.6-2.7 | PED、PES 与销售收入 | Elasticity and revenue

PED = percentage change in quantity demanded $\div$ percentage change in price. 绝对值大于 1 为 elastic, 小于 1 为 inelastic, 等于 1 为 unit elastic。对缺乏弹性的需求, 提高价格会使 total revenue

上升; 对富有弹性的需求, 提高价格会使 total revenue 下降。PES 取决于闲置产能、库存、生产周期和要素流动性。

English explanation. Price elasticity measures responsiveness, not the direction of a curve shift. Always show percentage changes and interpret the value in context. A firm facing elastic demand may raise revenue by lowering price.

知识点 01/02 | PED、PES 与销售收入: 弹性衡量“反应有多大” | Elasticity measures the size of a response

PED 使用百分比而非绝对变化, 使不同货币和数量单位可以比较。通常取绝对值分类: $|PED| > 1$ elastic, $|PED| < 1$ inelastic。必需品、替代品少、支出占比低以及短期通常使需求较缺乏弹性; 奢侈品、替代品多和较长时间使需求较富弹性。PES 则取决于 spare capacity、stocks、production period 和 factor mobility。

知识点 02/02 | PED、PES 与销售收入: PED 与 total revenue | Connecting PED with total revenue

$TR = P \times Q$ 。若需求缺乏弹性, 价格提高的百分比大于数量减少的百分比, TR 上升; 若需求富有弹性, 数量减少比例更大, TR 下降。计算时例如 P 由 10 增至 12 (20%), Q 由 100 降至 70 (30%), $PED = 30/20 = 1.5$, 属 elastic, TR 从 1000 降至 840。

PED 衡量 consumers 的 quantity demanded 对 price 的反应; PES 衡量 producers 的 quantity supplied 对 price 的反应。两者都使用 percentage change, 但决定因素与应用对象不同。

many close substitutes $\rightarrow$ consumers can switch easily after price rise $\rightarrow$ large percentage fall in $Q_d \rightarrow$ PED more elastic. spare capacity $\rightarrow$ firms expand quickly after price rise $\rightarrow$ larger percentage rise in $Q_s \rightarrow$ PES more elastic。

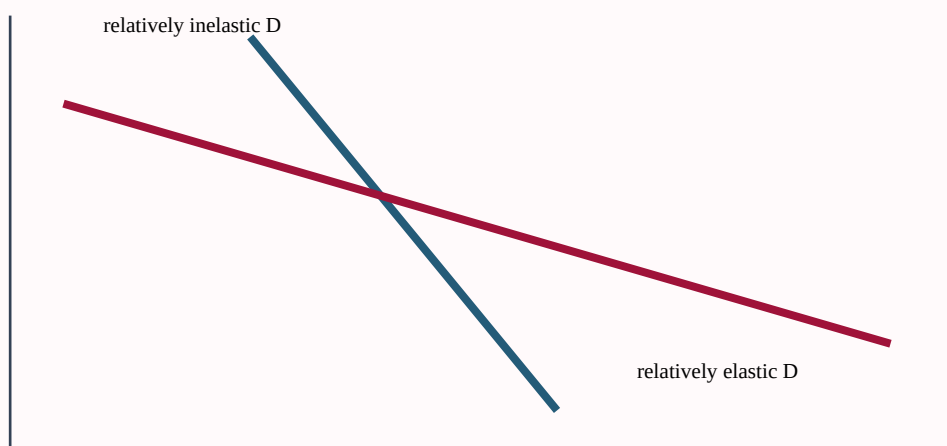
价格由 \$20 降至 \$18, 下降 10%; Q_d 由 200 增至 250, 增加 25%; $|PED| = 25/10 = 2.5$, elastic。TR 由 4000 增至 4500, 符合降价提高富弹性需求下收入。

弹性不是单看曲线视觉斜率。先写百分比变化公式, 再结合 substitutes、time period、necessity 和 capacity 解释数值。收入题还要把 price、quantity 与 $total\ revenue = P \times Q$ 连接。

Elasticity measures responsiveness, not slope alone. The same straight demand curve has different elasticity at different points. Determinants include substitutes, necessity, budget share and time; for supply, capacity, stock and production time are central.

弹性没有货币单位。不要把“2”解释为数量增加2个; 应写 quantity changes by about 2% for a 1% price change。

企业收入结论要求其他条件不变。竞争者反应、品牌差异和成本变化可能让实际 revenue 与简单 PED 预测不同。



图解：相同价格百分比变化下，数量反应较大的需求更有弹性；弹性必须用百分比计算，而不能只凭曲线外观。

2.9-2.10 | 市场失灵与政府干预 | Market failure and government intervention

市场失灵指自由市场没有把资源配置到社会最优位置，原因包括 external costs and benefits、public goods、information failure 和 market power。政策包括 indirect taxes、subsidies、regulation、information provision、price controls 和 direct provision。

English explanation. Market failure occurs when the price mechanism does not allocate resources efficiently. A tax may reduce consumption of a demerit good, but effectiveness depends on PED, enforcement costs and unintended effects.

知识点 01/02 | 市场失灵与政府干预：从私人决定到社会结果 | From private decisions to social outcomes

市场失灵发生在 free market allocation 不能最大化社会福利。负外部性中，生产者或消费者只考虑 private cost，却把污染、噪声或健康损害转嫁给 third parties，因此 social cost > private cost，市场数量可能过多。正外部性中 social benefit > private benefit，教育或疫苗可能被少消费。

知识点 02/02 | 市场失灵与政府干预：不同失灵需要不同政策 | Matching intervention to the source of market failure

Public goods 具有 non-excludability 和 non-rivalry, free-rider problem

使私人企业难收费，政府可能直接提供。Information failure 使消费者低估有益品或高估有害品。Indirect tax、subsidy、regulation、information campaign、maximum/minimum price 和 direct provision 各有成本、时滞与副作用。

Private decision-makers 依据 private costs and benefits；社会评价还包括 external costs and benefits。若市场价格没有反映第三方影响，market quantity 可能偏离 socially efficient quantity。

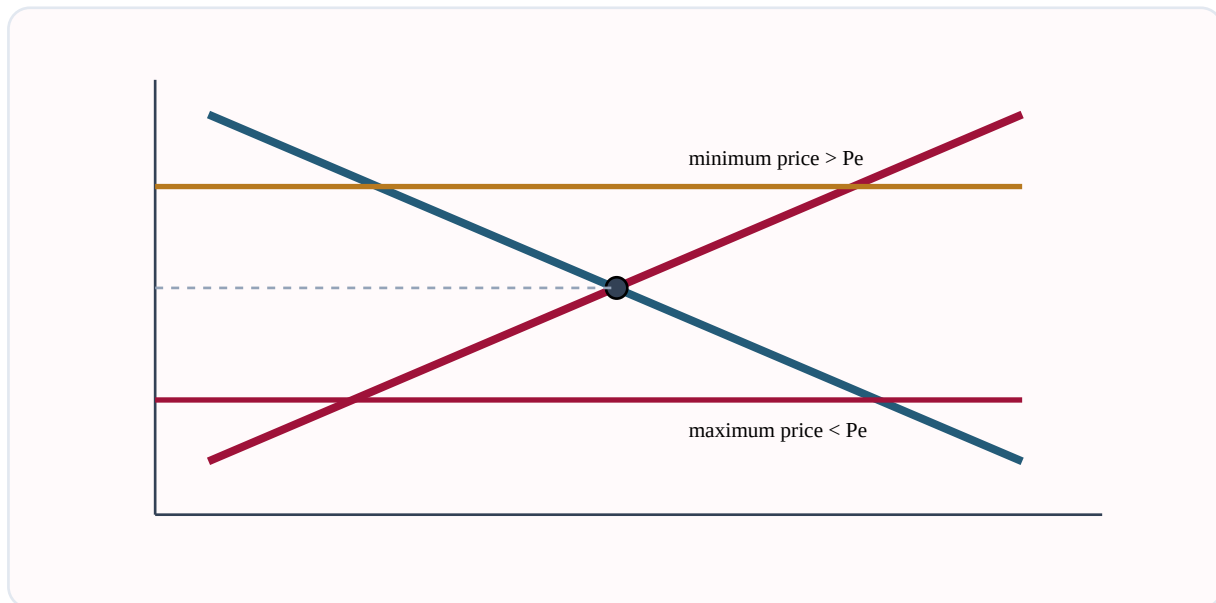
factory pollution → external cost imposed on residents → private cost below social cost → market price too low/output too high → indirect tax raises private cost → supply left → output and pollution may fall。

每单位产品 private cost=\$6，污染 external cost=\$3，则 social cost=\$9。若市场只按 \$6 作决定，会忽略 \$3 的第三方损害；理论上每单位 \$3 的税可使决策者面对更完整成本。

An externality affects a third party not directly involved in the transaction. Public goods are non-excludable and non-rival, creating a free-rider problem. Information failure occurs when choices are based on inaccurate or incomplete information.

Public good 不等于“政府提供的好东西”。必须同时具有 non-excludability 与 non-rivalry。

政府干预也有 information、administrative cost、unintended consequence 和 political pressure，评价时比较 market failure 与 government failure。



图解：最高限价必须低于 P_e 才造成 shortage；最低限价必须高于 P_e 才造成 surplus。

2.8-2.10 | 市场经济、计划经济与混合经济 | Economic systems and the mixed economy

Market economy 主要由价格和利润信号配置资源；planned economy

由政府决定主要生产什么、如何生产、为谁生产；现实多是 mixed economy（混合经济）。评价制度不能只列优缺点：市场激励创新但可能出现不平等、公共品不足和外部性；政府可纠正失灵但也不可能信息不足、官僚低效。

知识点 01/02 | 经济制度与资源配置：三种制度回答同样的三个问题 | How three systems answer the same economic questions

所有经济体都必须决定 what to produce, how to produce and for whom to produce。市场经济主要依靠价格、利润和消费者选择；计划经济主要依靠政府指令和公共所有；混合经济同时使用市场与政府。判断制度时不要看“有没有政府”或“有没有企业”，而要看关键资源配置决定由谁作出。

知识点 02/02 | 经济制度与资源配置：比较激励、信息、效率与公平 | Comparing incentives, information, efficiency and equity

价格上涨向企业发出 scarcity 和 profitability

信号，可能吸引资源进入该市场；竞争鼓励降成本和创新。然而市场可能少供 public goods、忽略 external costs，并依据购买力而非需要分配产品。政府可以提供医疗教育、征税或监管，但也不可能缺乏地方信息、反应缓慢或受到政治目标影响。现实结论通常取决于市场竞争程度、政府能力和产品性质。

Consumers、private firms 与 government 是三类关键决策者。市场制度主要通过 prices and profit 配置资源，计划制度主要通过 state instructions，mixed economy 则按产品性质组合两种机制。

消费者对太阳能设备需求增加 $\rightarrow$ price/profit signal strengthens $\rightarrow$ firms enter and hire resources $\rightarrow$ supply expands $\rightarrow$ resources shift away from less profitable uses。计划经济则可能由政府根据能源目标直接配置资源。

A market system can process dispersed information through prices, while planning can direct resources towards social priorities. Neither system guarantees efficiency or fairness. A mixed economy attempts to combine market incentives with public provision and regulation.

最佳组合取决于产品性质与政府能力。公共品、外部性强或信息不对称严重的领域通常需要更多干预，普通消费品更适合竞争。

医疗由市场提供可带来选择与创新，但 willingness and ability to pay 不等于 medical need。评价时把效率、信息不对称、外部收益、税收机会成本与政府执行能力放在同一结论中。

2027-2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 市场失灵的完整边界 | The scope of market failure

市场失灵是自由市场未能有效配置资源。原因包括 external costs/benefits、公共物品不被提供、merit goods 消费不足、demerit goods 消费过多，以及 monopoly 限制产量并提高价格。2027-2029 考纲要求理解这些机制，但不要求绘制市场失灵的社会成本-社会收益图；答题重点应放在清楚的文字因果链。

English explanation. Market failure occurs when the free market does not allocate resources efficiently. A public good may be underprovided because non-payers cannot easily be excluded. A monopoly may restrict output and charge a higher price because competitive pressure is weak. The syllabus tests these mechanisms through explanation and application rather than requiring market-failure diagrams.

补充知识点 02/02 | 价格管制、税、补贴与数量限制 | Government intervention

maximum price 设在均衡价以下才有约束力，并造成 shortage；minimum price 设在均衡价以上才有约束力，并造成 surplus。间接税提高企业单位成本，使供给左移；补贴降低有效成本，使供给右移。政府还可以 regulation、direct provision、nationalisation、privatisation 和 quota 干预。每项政策都要比较预期收益、财政或执行成本、弹性和 unintended consequences。

English explanation. A binding maximum price is below equilibrium and creates excess demand. A binding minimum price is above equilibrium and creates excess supply. An indirect tax shifts supply to the left, while a subsidy shifts it to the right. The size of the price and quantity changes depends partly on demand and supply elasticities.

3. 核心词汇表 | Core vocabulary (共 62 词)

English term	中文准确口径	Exam-ready meaning
market	市场	Any arrangement that brings buyers and sellers into contact.
buyer	买方	A person or organisation willing and able to purchase a product.
seller	卖方	A person or firm offering a product for sale.
consumer	消费者	A person who uses goods and services to satisfy wants.
firm	企业	An organisation that produces goods or services.
price mechanism	价格机制	The way price changes allocate resources through demand and supply.
demand	需求	The willingness and ability to buy a product at different prices over a period.
quantity demanded	需求量	The amount consumers plan to buy at a particular price.
individual demand	个人需求	The quantities one consumer plans to buy at different prices.
market demand	市场需求	The sum of all individual demand at each price.
demand schedule	需求表	A table showing price and quantity demanded.
ceteris paribus	其他条件不变	The assumption that relevant other factors remain unchanged.
extension in demand	需求量扩张	A rise in quantity demanded caused by a fall in the product's own price.

English term	中文准确口径	Exam-ready meaning
contraction in demand	需求量收缩	A fall in quantity demanded caused by a rise in the product's own price.
increase in demand	需求增加	A rightward shift of demand caused by a non-price factor.
decrease in demand	需求减少	A leftward shift of demand caused by a non-price factor.
substitute	替代品	A product that can be used instead of another product.
complement	互补品	A product used together with another product.
normal good	正常品	A product whose demand rises when income rises.
inferior good	低档品	A product whose demand falls when income rises.
supply	供给	The willingness and ability of producers to sell at different prices over a period.
quantity supplied	供给量	The amount producers plan to sell at a particular price.
extension in supply	供给量扩张	A rise in quantity supplied caused by a rise in own price.
contraction in supply	供给量收缩	A fall in quantity supplied caused by a fall in own price.
indirect tax	间接税	A tax on spending or production that raises firms' costs.
subsidy	补贴	A payment to producers that lowers effective production cost.
cost of production	生产成本	The expenditure incurred in producing output.
signalling function	信号功能	The use of price changes to communicate relative scarcity.
incentive function	激励功能	The way price and profit encourage buyers or sellers to change behaviour.
rationing function	配给功能	The allocation of scarce products through willingness and ability to pay.
equilibrium	均衡	A market position where quantity demanded equals quantity supplied.
shortage	短缺	Excess demand: quantity demanded is greater than quantity supplied.
surplus	过剩	Excess supply: quantity supplied is greater than quantity demanded.
upward pressure on price	价格上涨压力	Market pressure created when buyers compete for insufficient supply.
equilibrium price	均衡价格	The price at which quantity demanded equals quantity supplied.
equilibrium quantity	均衡数量	The quantity traded when the market is in equilibrium.
simultaneous shifts	供需同时移动	Changes in demand and supply occurring at the same time.
relative magnitude	相对幅度	The size of one curve shift compared with another.
price elasticity of demand (PED)	需求价格弹性	Responsiveness of quantity demanded to a change in price.
price elasticity of supply (PES)	供给价格弹性	Responsiveness of quantity supplied to a change in price.
elastic demand	富有弹性的需求	Demand with a proportionately larger quantity response than the price change.

English term	中文准确口径	Exam-ready meaning
inelastic demand	缺乏弹性的需求	Demand with a proportionately smaller quantity response than the price change.
total revenue	总收益	Price multiplied by quantity sold.
necessity	必需品	A product consumers find difficult to do without.
spare capacity	闲置生产能力	Unused resources that allow output to rise quickly.
market failure	市场失灵	A situation in which the market does not allocate resources efficiently.
externality	外部性	A cost or benefit affecting a third party outside a transaction.
private cost	私人成本	A cost paid by the decision-maker.
social cost	社会成本	Private cost plus external cost.
public good	公共品	A non-rival and non-excludable product.
merit good	有益品	A product that is under-consumed if left to the market.
demerit good	有害品	A product that is over-consumed if left to the market.
information failure	信息失灵	A lack of accurate information leading to poor decisions.
market economy	市场经济	An economy in which prices and private decisions allocate most resources.
planned economy	计划经济	An economy in which the state directs most resource allocation.
mixed economy	混合经济	An economy combining market decisions with government intervention.
consumer sovereignty	消费者主权	The influence consumers have over production through their spending.
private sector	私营部门	Economic activity owned and controlled by individuals or firms.
public sector	公共部门	Economic activity owned or controlled by the government.
equity	公平	A judgement about whether economic outcomes are fair.
maximum price	最高限价	A legal price ceiling, binding when set below equilibrium.
minimum price	最低限价	A legal price floor, binding when set above equilibrium.

4. 单元练习 | Unit practice (共 35 题)

练习 01/35

Question. Which is a market: only a physical shop, only a stock exchange, any arrangement connecting buyers and sellers, or any government office?

练习 02/35

Question. Explain two ways prices influence buyers and sellers in a market.

练习 03/35

Question. At a price of \$5, three consumers demand 4, 7 and 9 units. What is market quantity demanded?

练习 04/35

Question. Explain why a demand curve usually slopes downwards.

练习 05/35

Question. At a price of \$8, Ana demands 3 units, Ben 5 and Chen 4. Calculate market demand and state what must be held constant when adding the figures. [3]

练习 06/35

Question. The price of petrol rises. What happens to demand for petrol cars, assuming they are complements?

练习 07/35

Question. Analyse how higher household income may affect demand for bus travel.

练习 08/35

Question. Draw diagrams to distinguish a fall in the price of tea from a rise in the price of coffee, a substitute for tea. [6]

练习 09/35

Question. Which change shifts the supply of bread left: lower flour prices, a subsidy, improved ovens, or a tax on each loaf?

练习 10/35

Question. Explain two reasons why the supply of agricultural products may be price inelastic in the short run.

练习 11/35

Question. Draw and explain the effect of a subsidy paid to bicycle producers. [4]

练习 12/35

Question. At the current price, firms plan to sell 1 200 units while consumers plan to buy 850 units. Which outcome is most likely in a free market?

A Demand increases. B Price falls. C Supply shifts left. D Price rises.

练习 13/35

Question. The market price is below equilibrium. Which sequence correctly describes the adjustment?

A surplus → price rises; B shortage → price rises; C shortage → demand shifts right; D surplus → supply shifts left.

练习 14/35

Question. At \$5, Qd is 160 and Qs is 90. At \$8, Qd is 110 and Qs is 140. Identify and calculate the disequilibrium at each price.

练习 15/35

Question. Explain how the price mechanism may remove a shortage of bottled water. [4]

练习 16/35

Question. Analyse how a rise in the price of solar panels can perform the signalling, incentive and rationing functions. [6]

练习 17/35

Question. Discuss whether the price mechanism will always allocate essential medicines efficiently. [8]

练习 18/35

Question. Draw a demand and supply diagram showing a price above equilibrium and explain the adjustment.

练习 19/35

Question. Explain two reasons why a shortage may persist even when there is upward pressure on price. [4]

练习 20/35

Question. Demand falls while supply rises. Which outcome is certain?

练习 21/35

Question. Analyse the effect on the market for electric cars of a successful advertising campaign and a fall in battery costs.

练习 22/35

Question. Demand and supply both increase. Which equilibrium result is certain, and which is uncertain? Use a diagram to explain. [5]

练习 23/35

Question. Price rises by 8% and quantity demanded falls by 4%. What is PED and what happens to total revenue?

练习 24/35

Question. Analyse why demand for a life-saving medicine may be price inelastic and how this affects a tax.

练习 25/35

Question. Price falls by 10% and quantity demanded rises by 25%. Calculate PED and explain what happens to total revenue. [4]

练习 26/35

Question. Which is the clearest example of an external cost: a firm pays wages, a smoker harms nearby non-smokers, a consumer buys food, or a bank charges interest?

练习 27/35

Question. Discuss whether a tax is the best way to reduce consumption of sugary drinks.

练习 28/35

Question. Which feature is most characteristic of a market economy: consumer sovereignty, state production targets, equal income, or free public goods?

练习 29/35

Question. Discuss whether healthcare should be allocated entirely by the market.

练习 30/35

Question. Income rises and demand for bus journeys falls. Which description is correct?

A buses are normal goods B buses are inferior goods C quantity demanded has contracted D supply has decreased

练习 31/35

Question. At a price of \$12, $Q_d = 180$ and $Q_s = 240$. What market pressure exists?

A shortage of 60 and upward pressure B shortage of 420 and downward pressure C surplus of 60 and downward pressure D surplus of 420 and upward pressure

练习 32/35

Question. The price of a product rises from \$20 to \$22 and quantity demanded falls from 500 to 450. Using percentage changes based on the original values, calculate PED and classify demand. [4]

练习 33/35

Question. Explain why a maximum price for rented housing may create a shortage. [4]

练习 34/35

Question. Analyse how a subsidy for solar panels may correct market failure. [6]

练习 35/35

Question. Discuss whether an indirect tax will greatly reduce cigarette consumption. [8]

5. 练习答案与解析 | Answers and explanations (共 35 题)

答案 01/35

正确答案 | Correct answer. Any arrangement connecting buyers and sellers. Physical location is not required.

答案 02/35

参考答案 | Model answer. A higher price rations the product by reducing the quantity consumers are willing and able to buy. It also provides sellers with an incentive to increase quantity supplied because potential revenue and profit rise. These responses help coordinate the two sides of the market.

答案 03/35

正确答案 | Correct answer. 20 units. Market demand is the horizontal sum of individual quantities at the same price.

答案 04/35

参考答案 | Model answer. When price falls, the product becomes cheaper relative to substitutes, so consumers may switch towards it. The lower price also increases real purchasing power, enabling consumers to buy more. Hence quantity demanded rises, *ceteris paribus*.

答案 05/35

正确答案 | Correct answer. Market quantity demanded is $3 + 5 + 4 = 12$ units. The price, time period and product unit must be the same for every consumer; prices are not added.

答案 06/35

正确答案 | Correct answer. Demand for petrol cars decreases and its demand curve shifts left. The car price need not have changed.

答案 07/35

参考答案 | Model answer. If bus travel is an inferior service, higher income enables households to switch to cars or taxis. Demand for bus travel falls and its curve shifts left. If buses are high-quality and environmentally preferred, they may instead be normal, so the result depends on consumer behaviour.

答案 08/35

正确答案 | Correct answer. For the fall in tea price, use one tea demand curve and show a downward movement to a larger quantity demanded. For the rise in coffee price, keep tea price unchanged and shift the whole tea demand curve right because consumers switch from coffee to tea.

答案 09/35

正确答案 | Correct answer. A tax on each loaf, because it raises unit cost and reduces profitability at every market price.

答案 10/35

参考答案 | Model answer. Crops require time to grow, so farmers cannot quickly raise current harvests when price rises. Available land and storage may also be limited. Hence the percentage change in quantity supplied is smaller than the percentage change in price.

答案 11/35

正确答案 | Correct answer. Label Price and Quantity, draw D and S1, then shift supply right to S2. A subsidy lowers the effective cost of production and increases the profit obtainable at each market price, so producers are willing and able to supply more.

答案 12/35

正确答案 | Correct answer: B – Price falls. Qs exceeds Qd by 350 units, so unsold stocks accumulate. Sellers have an incentive to reduce price. The price fall causes an extension in quantity demanded and a contraction in quantity supplied; neither curve has to shift.

答案 13/35

正确答案 | Correct answer: B – shortage → price rises. Below equilibrium, $Q_d > Q_s$. Competition among buyers produces upward pressure on price. Higher price contracts quantity demanded and extends quantity supplied until $Q_d = Q_s$.

答案 14/35

正确答案 | Correct answer. At \$5 there is a shortage of 70 units because $160 - 90 = 70$. At \$8 there is a surplus of 30 units because $140 - 110 = 30$. The \$5 price faces upward pressure; the \$8 price faces downward pressure.

答案 15/35

参考答案 | Model answer. A shortage means quantity demanded is greater than quantity supplied at the current price [1]. Buyers compete for limited bottles, creating upward pressure on price [1]. The higher price causes a

contraction in quantity demanded and an extension in quantity supplied [1]. Adjustment continues until Q_d equals Q_s at the equilibrium price [1].

答案 16/35

参考答案 | Model answer. The higher price signals that panels have become relatively scarce or that demand has strengthened [1]. It tells firms that potential revenue and profit may be higher [1]. This gives existing producers an incentive to expand and may attract new firms, increasing supply over time [1–2]. The higher price also reduces quantity demanded [1], rationing the limited panels to consumers willing and able to pay [1].

答案 17/35

参考答案 | Model answer. Prices can signal scarcity, encourage production and ration limited stocks. Profit can attract investment and innovation. However, willingness and ability to pay may not reflect medical need; low-income patients can be excluded. Patents, imperfect information and market power may keep prices high, while positive external benefits from treatment may cause under-consumption. A justified conclusion is that markets can coordinate ordinary supply, but subsidy, regulation or direct provision may be needed for essential medicines. The best policy depends on fiscal cost, targeting and government information.

答案 18/35

正确答案 | Correct answer. The diagram must label Price, Quantity, D, S, E, P_e and a higher P_1 . At P_1 , mark Q_d on the left and Q_s on the right; the horizontal gap is surplus. State $Q_s > Q_d \rightarrow$ surplus $\rightarrow$ unsold stocks $\rightarrow$ downward pressure on price $\rightarrow$ extension in demand and contraction in supply $\rightarrow$ equilibrium.

答案 19/35

参考答案 | Model answer. A government may impose a maximum price below equilibrium, preventing the legal price from rising; quantity demanded therefore remains above quantity supplied. Alternatively, long contracts or very low short-run PES may delay producers' response, so supply cannot increase quickly. Queues, rationing or a black market may then appear. Each reason must identify the obstacle and connect it to the continuing gap between Q_d and Q_s .

答案 20/35

正确答案 | Correct answer. Equilibrium price falls. Both shifts put downward pressure on price, while their effects on quantity oppose each other.

答案 21/35

参考答案 | Model answer. Advertising shifts demand right. Lower battery costs shift supply right. Both changes increase equilibrium quantity. Demand growth raises price while supply growth lowers it, so the final price depends on the relative size of the shifts.

答案 22/35

正确答案 | Correct answer. Shift D and S to the right. Equilibrium quantity certainly rises because both changes raise quantity. Equilibrium price is uncertain because the demand increase raises price while the supply increase lowers it; the final direction depends on the relative size of the shifts.

答案 23/35

正确答案 | Correct answer. PED is 0.5 in absolute value, so demand is inelastic and total revenue is likely to rise.

答案 24/35

参考答案 | Model answer. The medicine is a necessity and may have few substitutes, so patients reduce purchases by a smaller percentage than the price rise. A tax can therefore raise substantial revenue, but consumption and the related external cost may fall only slightly. Low-income patients may bear a large burden.

答案 25/35

正确答案 | Correct answer. $PED = 25\% / -10\% = -2.5$, or 2.5 in absolute value. Demand is price-elastic. Quantity rises proportionately more than price falls, so total revenue rises, other things equal.

答案 26/35

正确答案 | Correct answer. Harm to nearby non-smokers, because the cost falls on third parties outside the transaction.

答案 27/35

参考答案 | Model answer. A tax raises firms' costs and price, reducing quantity demanded and health external costs while raising revenue. However, demand may be inelastic, the tax may be regressive and substitutes may also be unhealthy. Information campaigns or regulation may complement the tax. Effectiveness depends on tax size and alternatives.

答案 28/35

正确答案 | Correct answer. Consumer sovereignty, because purchases influence what profit-seeking firms produce.

答案 29/35

参考答案 | Model answer. Market provision may encourage efficiency, choice and innovation. Competition can reduce cost. However, low-income patients may be unable to pay, consumers may lack medical information and healthcare creates external benefits. Public funding or regulation may improve access, although taxation has opportunity cost and government provision may be inefficient.

答案 30/35

正确答案 | Correct answer. B – Demand changes because income, a non-price determinant, has changed. Falling demand as income rises identifies bus journeys in this case as an inferior good.

答案 31/35

正确答案 | Correct answer. $C - Q_s - Q_d = 240 - 180 = 60$, so unsold stocks create downward pressure on price.

答案 32/35

正确答案 | Correct answer. $\% \Delta Q_d = -50/500 \times 100 = -10\%$. $\% \Delta P = 2/20 \times 100 = 10\%$. $PED = -10\%/10\% = -1$, so demand is unit elastic. If the convention uses the absolute value, $PED = 1$.

答案 33/35

参考答案 | Model answer. If the maximum price is set below equilibrium, rent is lower. This causes an extension in quantity demanded but a contraction in quantity supplied along the existing curves. Therefore Q_d exceeds Q_s and a shortage develops. Some tenants may face waiting lists or non-price rationing.

答案 34/35

参考答案 | Model answer. A subsidy lowers producers' effective cost, shifting supply right. The market price falls and quantity bought rises. If solar panels create external benefits by reducing pollution, the free market may

otherwise under-consume them because consumers consider mainly private benefits. Higher consumption can move output closer to the socially desirable level, although the policy has an opportunity cost for government.

答案 35/35

参考答案 | Model answer. The tax shifts supply left, raising price and reducing quantity. It may internalise external costs and raise revenue. However, cigarette demand may be price inelastic because of addiction and few close substitutes, so quantity falls by a smaller percentage than price rises. Effectiveness may increase over time as users adjust, and it may improve if combined with information, age restrictions and cessation support. A high tax can also encourage smuggling and be regressive.

6. 自学检查清单 | Self-study checklist (共 12 项)

- ☐ 能写出 demand 与 supply 的完整定义
- ☐ 能按同一价格横向加总 market demand 与 market supply
- ☐ 能区分 extension/contraction 与 increase/decrease in demand or supply
- ☐ 能从非价格因素判断曲线移位方向并解释因果链
- ☐ 能计算 shortage、surplus 并判断价格压力
- ☐ 能解释价格机制的 signalling、incentive、rationing functions
- ☐ 能处理需求与供给同时变化时的确定结论和不确定结论
- ☐ 能计算、分类并解释 PED 与 PES
- ☐ 能用 PED 判断价格变化对 total revenue 的影响
- ☐ 能分析 public、merit、demerit goods、externalities 和 monopoly
- ☐ 能正确画 maximum/minimum price、indirect tax 和 subsidy
- ☐ 能评价政府干预的收益、成本、弹性与时间条件

单元 3 | 微观经济决策者 | Microeconomic decision makers

官方范围 | Official scope: 3.1–3.7

1. 单元导读 | Unit introduction

本单元把市场模型中的“消费者”和“生产者”拆成具体决策者：家庭决定消费、储蓄与借款；劳动者选择职业；银行连接储蓄与投资；企业选择要素、规模、产量和目标。计算题集中在 productivity、cost、revenue 和 profit，解释题则要求把利率、工资、成本和激励连成链。

与人教版衔接及口径差异 | Connection with Chinese curricula. 与人教版货币、劳动就业和企业经营衔接。Cambridge 更强调 money 的四项功能、中央银行与商业银行的分工、labour demand 的派生性质，以及固定/可变/平均成本之间的精确计算。中文“资本”可能指资金，Cambridge 的 factor of production capital 通常指 produced means of production。

学习路线 | Learning route.

1. 3.1 货币、中央银行与商业银行 – Money and banking
2. 3.2–3.3 家庭、劳动者、工资与工会 – Households, workers, wages and trade unions
3. 3.4–3.6 企业生产、生产率、成本与收益 – Production, productivity, costs and revenue
4. 3.4, 3.7 企业规模、合并与市场结构 – Firm size, integration and market structure
5. 3.5–3.6 从成本表到企业决策 – Using cost data in firm decisions

2. 知识点精讲 | Detailed teaching

3.1 | 货币、中央银行与商业银行 | Money and banking

货币的四项功能是 medium of exchange（交换媒介）、measure of value（价值尺度）、store of value（价值储藏）与 standard of deferred payment（延期支付标准）。商业银行（commercial bank）接受存款、提供支付服务并放贷；中央银行（central bank）通常发行货币、监管银行并影响利率。通胀削弱货币的购买力，因此“钱可以储藏价值”不等于“价值永远不变”。

知识点 01/02 | 货币与银行：货币功能与良好货币特征 | Functions and characteristics of money

货币作为 medium of exchange 消除 barter 的 double coincidence of wants；作为 measure of value 让商品可比较；作为 store of value 把购买力转移到未来；作为 standard of deferred payment 支持借贷。良好货币应 durable、portable、divisible、recognisable、scarce and stable in value。高通胀会削弱价值储藏功能。

知识点 02/02 | 货币与银行：商业银行与中央银行 | Commercial-bank and central-bank roles

商业银行接受 deposits、提供 loans、支付和外汇服务，并通过利差盈利。中央银行通常发行货币、监管银行、作为政府银行和最后贷款人，并实施 monetary policy。利率上升提高借款成本，可能减少消费和投资；同时提高储蓄回报。结果取决于信心、债务类型和利率变化幅度。Households 持有货币和存款，commercial banks 接受存款并提供贷款与支付，central bank 管理货币体系与利率。必须区分 money functions、bank functions 与 interest-rate effects。central bank raises interest rate → borrowing becomes dearer/saving more attractive → household consumption and firm investment may fall → AD falls → inflationary pressure may ease。

存款 \$1 000、年利率3%，一年利息约 \$30。若通胀5%，实际购买力仍可能下降约2%，说明 nominal return 与 real return 不同。

Money performs four functions: medium of exchange, measure of value, store of value and standard of deferred payment. Commercial banks accept deposits and make loans. The central bank conducts monetary policy, supervises the banking system and may act as lender of last resort.

中央银行通常不为普通家庭办理日常存款；商业银行也不负责制定全国政策利率。

3.2-3.3 | 家庭、劳动者、工资与工会 | Households, workers, wages and trade unions

家庭提供 labour、land、capital，获得收入后在消费与储蓄之间选择。劳动需求是 derived demand（派生需求）：企业需要工人，是因为消费者需要工人生产的商品。工资、技能、资格、工作条件、歧视和工会谈判力量都会影响工资差异。工会可用集体谈判提高工资和安全，也可能在工资高于均衡时增加失业；要写“可能”，不能绝对化。

知识点 01/02 | 家庭、劳动者与工会：家庭如何分配收入 | How households allocate income

家庭收入可用于 consumption、saving 和 direct

taxes。收入越高，通常能储蓄的绝对额越大，但比例还受利率、信心、年龄、财富和未来预期影响。消费者选择也受 price、income、advertising、quality 和 tastes 影响。借贷让当前消费增加，但未来必须偿还本息。

知识点 02/02 | 家庭、劳动者与工会：工资差异与劳动市场 | Wage differences and the labour market

工资由劳动需求与供给共同影响。需要高技能、培训期长、风险大或工作条件差的职业可能工资较高；工会、最低工资、歧视和公共部门政策也会改变工资。Trade union 通过 collective bargaining 改善工资和条件，但工资高于生产率增长可能增加成本、价格或失业风险。

Households 同时是 consumers、savers 与 labour suppliers；workers 关注 wage、conditions 和 job security；trade unions 通过 collective bargaining 改变谈判力量。

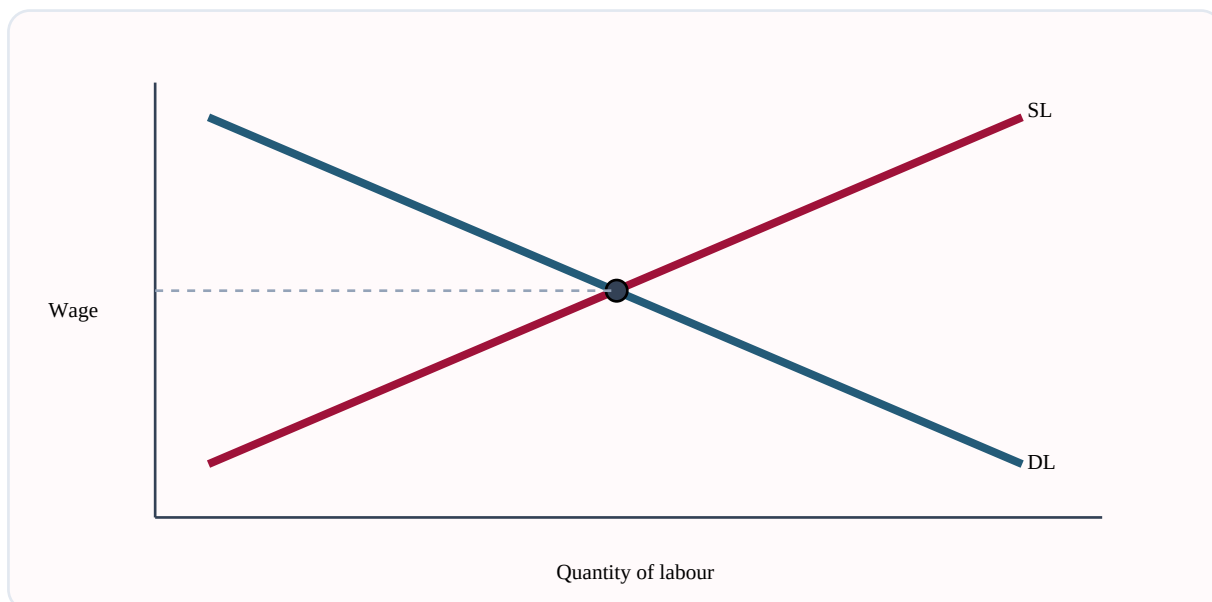
product demand rises → derived demand for labour rises → labour shortage strengthens worker bargaining power → wages may rise → firms' costs may increase, unless productivity also rises.

月收入 \$2 000，direct tax \$300，消费 \$1 400，则 saving=\$300。利率上升不保证储蓄必然增加：income effect 与 substitution effect 可能方向不同。

Labour demand is derived because firms employ workers to produce goods and services. Workers compare wages with hours, conditions, security, promotion and location. Trade unions use collective bargaining and may also provide legal or training services.

Labour supply increase 是更多工人愿意提供劳动；employment increase 是实际受雇人数增加，两者不是同义词。

工资高低还应与 real wage、工作时长、生活成本、福利和就业稳定性一起判断。



图解：劳动需求与劳动供给的交点决定均衡工资和就业量；有约束力的最低工资在均衡工资以上。

3.4–3.6 | 企业生产、生产率、成本与收益 | Production, productivity, costs and revenue

Fixed costs (租金、厂房) 不随短期产量变化; variable costs (原料、按件工资) 会变化。Total cost = fixed cost + variable cost。Total revenue = price × quantity; profit = total revenue – total cost。企业不只追求利润, 也可能追求 sales revenue、market share、survival 或环境声誉; 题目给了目标才应据此分析。

知识点 01/02 | 企业的生产、成本与收益: 生产率、分工与生产阶段 | Productivity, specialisation and stages of production

生产率是单位投入产出, 例如 output per worker, 不等于总产量。Division of labour 把生产拆成任务, 使工人 specialised, 减少转换时间并便于使用机器; 但重复工作可能降低 motivation, 停工也会影响整条生产线。Primary、secondary、tertiary sectors 分别涉及资源提取、制造建筑和服务。

知识点 02/02 | 企业的生产、成本与收益: 成本收益必须按公式连接 | Linking costs, revenue and profit with formulas

FC 不随 output 改变, VC 随 output 改变; $TC = FC + VC$, $AC = TC/Q$ 。TR = $P \times Q$, profit = $TR - TC$ 。若产量由 100 增至 200, TC 从 1000 增至 1600, AC 从 10 降至 8, 说明平均成本下降, 即使总成本上升。利润变化必须同时看 price、quantity 和 cost。

Firms 把 factors of production 转化为 output。Productivity 连接投入与产出; cost、revenue 与 profit 连接生产决定与财务结果; 短期至少一种要素固定, 长期所有要素可变。

division of labour → repeated practice and less switching time → output per worker rises → unit cost may fall → price may fall or profit rise → sales/investment may increase。

$Q=100$, $FC=\$400$, $VC=\$600$, 所以 $TC=\$1\,000$ 、 $AC=\$10$ 。若 price = $\$14$, $TR=\$1\,400$ 、profit = $\$400$ 。Q 增加后总成本可上升而平均成本下降。

Production is the transformation of inputs into output. Division of labour can raise productivity through specialisation, but repetitive work may reduce motivation. Profit is a residual: total revenue minus total cost, not the same as sales revenue.

FC 不随 output 改变, 但 per-unit fixed cost 会变; VC 随 output 变, 不代表每单位 VC 必须恒定。

Q	FC	VC	TC	AC
0	1200	1200	2400	–
50	1200	1800	3000	60.00
100	1200	2600	3800	38.00
150	1200	3600	4800	32.00

图解: $TC=FC+VC$; 平均成本把总成本分摊到每单位产量。必须分清 total 与 average。

3.4, 3.7 | 企业规模、合并与市场结构 | Firm size, integration and market structure

长期扩张时, 平均成本下降叫 economies of scale: 可以来自批量采购、专业化、管理、技术和融资。规模太大可能产生 communication problems 和 coordination problems, 即 diseconomies of scale。市场结构从竞争强到弱大致为 perfect competition、monopolistic competition、oligopoly、monopoly; 答题不要只背定义, 应把进入壁垒、产品差异、定价能力和消费者选择联系起来。

知识点 01/02 | 规模经济与市场结构: 企业变大为何可能先降成本 | Why expansion may initially lower average cost

Internal economies of scale 包括 purchasing、technical、financial、managerial、marketing 和 risk-bearing economies。大型企业批量采购可议价, 专用机器产量大时才划算, 借款风险较低, 专业经理提高决策质量, 广告成本可分摊到更多产品。它们影响 long-run average cost, 不等于短期固定成本分摊。

知识点 02/02 | 规模经济与市场结构: 规模不经济与竞争程度 | Diseconomies of scale and market competition

企业过大可能出现 communication failure、协调困难、管理层级增加和员工疏离, 使 average cost 上升。竞争市场中 firms 多、进入较易, 企业面临降价和创新压力; monopoly 有规模优势和研发资金, 也可能提高价格、限制选择或降低效率。市场份额高不自动证明滥用市场力量。

Internal economies 来自单个企业扩大, external economies 来自整个行业或地区扩大。市场结构则看 firms number、entry barriers、product differentiation 与 market power, 不能只用企业规模判断。

bulk purchasing $\rightarrow$ stronger bargaining power $\rightarrow$ lower input price per unit $\rightarrow$ average cost falls $\rightarrow$ firm can lower price $\rightarrow$ market share may rise $\rightarrow$ further scale economies may develop。

小企业年广告费 \$20 000、销量 10 000, 广告成本每件 \$2; 大企业广告费 \$100 000、销量 100 000, 每件 \$1。这是 marketing economy, 但总广告费更高。

Internal economies arise from the firm's own expansion; external economies arise when the industry or region expands. Diseconomies often result from communication, coordination and motivation problems in very large organisations.

规模经济是 average cost 下降, 不是 total cost 下降; market share 大也不等于只有一家企业。

Natural monopoly 中重复基础设施成本高, 受监管单一企业可能比许多小企业有效; 其他市场则竞争更可能保护消费者。

3.5–3.6 | 从成本表到企业决策 | Using cost data in firm decisions

把题中的每一列先翻译: TC 是总成本; $AC = TC/Q$; $TR = P \times Q$; profit = $TR-TC$ 。若固定成本不变, 产量上升时 AC 可能下降, 因为固定成本被更多单位分摊; 这不代表每一项成本下降。

例子: 厂房租金每月 10 000, 生产 1 000 件时每件固定成本 10; 生产 2 000 件时每件固定成本 5。若增加产量需要加班和更昂贵的原料, variable cost 仍可能上升。因此“规模扩大一定更便宜”错误。

写工会题时分两条线: 工资提高会提升会员收入; 企业劳动力成本提高可能减少雇佣、提高价格或投资自动化。结论取决于 labour demand elasticity、生产率和企业利润率。

知识点 01/02 | 从成本表到企业决策：读表顺序决定是否出错 | A reliable order for reading a cost table

先识别 output，再找固定成本是否在各产量相同；计算 $VC=TC-FC$ ；再计算 $AC=TC/output$ 、 $TR=price \times quantity$ 和 $profit=TR-TC$ 。零产量时 TC 通常等于 FC，但 AC 无法用 0 作除数。不要因为 AC 下降就写 TC 下降，也不要把 revenue 当 profit。

知识点 02/02 | 从成本表到企业决策：数据变化要解释原因 | Explaining why cost and revenue figures change

AC 下降可能来自固定成本分摊、specialisation 或更高 machinery utilisation；随后上升可能来自 overtime、拥堵和协调问题。利润最大并不一定发生在 output 最大处，因为额外产量也带来额外成本。IGCSE 通常不要求边际分析，但必须比较收益与成本。

FC 不随 output 变化，VC 随 output

变化， $TC=FC+VC$ ； $AC=TC/Q$ ， $MC=\Delta TC/\Delta Q$ ； $TR=P \times Q$ ， $profit=TR-TC$ 。每个公式回答不同问题。

$Q=0$ 时 $FC=240$ 、 $TC=240$ 。 $Q=40$ 时 $TC=600$ ，所以 $VC=360$ 、 $AC=15$ 。 $price=18$ 时 $TR=720$ 、 $profit=120$ 。若把 revenue 720 当 profit，会忽略全部成本。

零产量时 AC 不可计算，因为不能除以零；profit 可能为负，称 loss。

2027-2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 职业选择、工资差异与劳动力流动 | Occupational choice and labour mobility

职业选择受工资、工作条件、晋升、培训要求、风险、地点和个人偏好影响。工资差异可能来自技能与资格、劳动需求、劳动供给、生产率、工会力量、歧视或补偿性工资差异。劳动力不流动包括 occupational immobility 和 geographical immobility，会使失业与职位空缺同时存在。

English explanation. Workers do not choose occupations on wage alone. Non-wage factors include hours, safety, status, promotion and location. Wage differences can reflect differences in productivity, qualifications, demand and supply, bargaining power and working conditions. Labour immobility slows the movement of workers from declining to expanding industries.

补充知识点 02/02 | 企业合并与生产方法 | Integration and factor intensity

horizontal merger 发生在同一生产阶段，可能扩大市场份额和规模经济；vertical merger 连接上下游，可稳定原料或销售渠道；conglomerate merger 进入无直接关系的市场以分散风险。企业选择 labour-intensive 或 capital-intensive 生产，不仅比较工资与机器价格，还要比较生产率、产量规模、融资条件、质量和灵活性。

English explanation. A firm chooses a production method by comparing the productivity and cost of factor combinations. Capital-intensive production may lower unit cost at high output, but requires finance and may be inflexible. Labour-intensive production can adapt more easily and may suit customised services. Integration can reduce costs, but larger size may create communication and control problems.

3. 核心词汇表 | Core vocabulary (共 39 词)

English term	中文准确口径	Exam-ready meaning
money	货币	Anything generally accepted as payment for goods and services.
medium of exchange	交换媒介	A function of money that removes the need for barter.
store of value	价值储藏	A function allowing purchasing power to be held for later use.

English term	中文准确口径	Exam-ready meaning
commercial bank	商业银行	A bank accepting deposits and providing loans and payment services.
central bank	中央银行	The institution responsible for the monetary system and banking stability.
interest rate	利率	The price of borrowing and the reward for saving.
liquidity	流动性	How easily an asset can be converted into spendable money.
disposable income	可支配收入	Income available after direct taxes and benefits.
saving	储蓄	Income not spent on current consumption.
wage rate	工资率	Payment for labour per unit of time or output.
earnings	劳动收入	Total income received from employment.
trade union	工会	An organisation representing workers' interests.
collective bargaining	集体谈判	Negotiation between a union and an employer.
derived demand	派生需求	Demand for a factor arising from demand for the final product.
production	生产	The process of creating goods and services.
division of labour	劳动分工	Breaking production into tasks with workers specialising in particular tasks.
specialisation	专业化	Concentrating resources on a narrower range of activities.
fixed cost	固定成本	A cost that does not change with output in the short run.
variable cost	可变成本	A cost that changes as output changes.
average cost	平均成本	Total cost divided by output.
profit	利润	Total revenue minus total cost.
labour productivity	劳动生产率	Output per worker or per hour worked.
economies of scale	规模经济	Falls in long-run average cost as a firm expands.
diseconomies of scale	规模不经济	Rises in long-run average cost as a firm becomes too large.
internal economy	内部规模经济	A cost advantage arising from growth of the firm itself.
external economy	外部规模经济	A cost advantage arising from growth of the industry.
competition	竞争	Rivalry between firms for customers and sales.
monopoly	垄断	A market dominated by one supplier.
market share	市场份额	A firm's sales as a percentage of total market sales.
total cost (TC)	总成本	Fixed cost plus variable cost.
average total cost (AC)	平均总成本	Total cost divided by output.
total revenue (TR)	总收益	Price multiplied by quantity sold.
loss	亏损	A situation in which total cost exceeds total revenue.
profit margin	利润率	Profit expressed relative to sales revenue.
break-even	盈亏平衡	The position where total revenue equals total cost.
occupational mobility	职业流动性	The ability of labour to move between occupations.

English term	中文准确口径	Exam-ready meaning
geographical mobility	地域流动性	The ability of labour to move between locations.
horizontal integration	横向合并	A merger between firms at the same stage of production.
vertical integration	纵向合并	A merger between firms at different stages of production.

4. 单元练习 | Unit practice (共 17 题)

练习 01/17

Question. Which function of money is weakened most directly by rapid inflation?

练习 02/17

Question. Analyse how a rise in interest rates may affect households and firms.

练习 03/17

Question. Which change is most likely to increase demand for chefs: lower demand for restaurant meals, higher chef productivity, more people entering chef training, or lower food prices with no sales change?

练习 04/17

Question. Discuss whether trade-union action always benefits workers.

练习 05/17

Question. A firm's output doubles while total cost rises by 50%. What happens to average cost?

练习 06/17

Question. Analyse two ways improved labour productivity may affect a firm.

练习 07/17

Question. Which is an internal economy of scale: lower national interest rates, a skilled regional labour pool, bulk-buying discounts, or a new public road?

练习 08/17

Question. Discuss whether consumers benefit from a monopoly.

练习 09/17

Question. At output 50, total cost is \$900 and variable cost is \$650. What is fixed cost?

练习 10/17

Question. Analyse why a firm may continue producing even when total profit falls temporarily.

练习 11/17

Question. Output rises from 40 to 50 units and total cost rises from \$600 to \$720. Calculate marginal cost per extra unit and explain what it measures. [3]

练习 12/17

Question. Which is a function of a central bank rather than a commercial bank?

A accepting household deposits B issuing notes and coins C providing debit cards D making business loans

练习 13/17

Question. A firm has fixed cost \$1 200. At output 200 units, variable cost is \$2 800 and total revenue is \$5 000. Calculate total cost, average cost and profit. [5]

练习 14/17

Question. Ten workers produce 2 400 units in a 40-hour week. Calculate output per worker and output per worker-hour. [4]

练习 15/17

Question. Explain two reasons why a worker with more qualifications may earn a higher wage. [6]

练习 16/17

Question. Analyse why a firm may change from labour-intensive to capital-intensive production. [6]

练习 17/17

Question. Discuss whether consumers always benefit when two large firms merge. [8]

5. 练习答案与解析 | Answers and explanations (共 17 题)

答案 01/17

正确答案 | Correct answer. Store of value, because money loses purchasing power quickly over time.

答案 02/17

参考答案 | Model answer. Loan and mortgage repayments may rise, reducing disposable income and consumption. Saving becomes more attractive. Firms face a higher cost of borrowing, so fewer investment projects are profitable. Demand and employment may fall, although savers receive more interest.

答案 03/17

正确答案 | Correct answer. Higher chef productivity, because each chef contributes more output and revenue to the firm.

答案 04/17

参考答案 | Model answer. Collective bargaining may raise wages, improve safety and reduce exploitation. Higher pay can motivate workers and reduce turnover. However, large wage increases can raise costs, encourage automation or reduce employment. Non-members and consumers may lose. The result depends on productivity and labour-demand elasticity.

答案 05/17

正确答案 | Correct answer. Average cost falls, because cost rises less proportionately than output.

答案 06/17

参考答案 | Model answer. More output per worker can reduce unit labour cost, allowing a lower price and greater competitiveness. Alternatively, the firm may keep price unchanged and earn more profit, financing investment. The effect depends on wages, product quality and demand.

答案 07/17

正确答案 | Correct answer. Bulk-buying discounts, because they result directly from the firm's larger purchasing scale.

答案 08/17

参考答案 | Model answer. A monopoly may exploit scale economies, fund research and provide a standardised network. It may pass lower costs to consumers. Conversely, weak competition may permit higher prices, less choice and lower quality. Outcomes depend on regulation, contestability and the source of monopoly power.

答案 09/17

正确答案 | Correct answer. \$250. Fixed cost equals total cost minus variable cost.

答案 10/17

参考答案 | Model answer. It may expect demand to recover, wish to retain skilled workers or avoid high closure and restart costs. If revenue covers variable cost and contributes to fixed cost, short-run production can reduce losses. In the long run, persistent revenue below total cost is unsustainable.

答案 11/17

正确答案 | Correct answer. The extra cost is $\$720 - \$600 = \$120$ for 10 additional units, so marginal cost per extra unit is \$12. It measures the addition to total cost caused by producing one more unit over this interval.

答案 12/17

正确答案 | Correct answer. B – A central bank normally issues the currency and acts as banker to government and commercial banks. The other services are commonly provided by commercial banks.

答案 13/17

正确答案 | Correct answer. $TC = FC + VC = 1\,200 + 2\,800 = \$4\,000$. $AC = TC/Q = 4\,000/200 = \20 per unit. Profit = $TR - TC = 5\,000 - 4\,000 = \$1\,000$.

答案 14/17

正确答案 | Correct answer. Output per worker = $2\,400/10 = 240$ units per week. Total worker-hours = $10 \times 40 = 400$. Output per worker-hour = $2\,400/400 = 6$ units.

答案 15/17

参考答案 | Model answer. Qualifications may raise productivity, increasing the value of the worker's output and therefore firms' demand for that labour. Qualifications may also take time and money to acquire, restricting labour supply; with stronger demand and more limited supply, equilibrium wage may be higher.

答案 16/17

参考答案 | Model answer. If wages rise relative to machinery costs, capital becomes comparatively cheaper. Automation may also produce more consistent output and raise productivity, reducing unit cost at a large scale. The firm will compare finance costs, required skills, maintenance, product quality and flexibility before changing methods.

答案 17/17

参考答案 | Model answer. A merger may create purchasing, technical and managerial economies of scale, lowering average cost and possibly price. It may finance innovation and improve reliability. However, reduced competition can give the merged firm market power, allowing higher prices, less choice and weaker service. The result depends on the cost savings, ease of entry, strength of remaining rivals and regulation, and on whether savings are passed to consumers.

6. 自学检查清单 | Self-study checklist (共 12 项)

- ☐ 能解释货币四项功能并判断良好货币的特征
- ☐ 能区分 central bank 与 commercial bank
- ☐ 能分析收入、利率、信心、年龄和文化对消费储蓄借款的影响
- ☐ 能解释职业选择与工资差异
- ☐ 能画劳动市场并分析 minimum wage
- ☐ 能解释 labour mobility 与 division of labour 的收益和成本
- ☐ 能区分三大产业、公私部门及企业规模
- ☐ 能解释 horizontal、vertical、conglomerate integration
- ☐ 能计算 productivity、TC、AC、FC、VC、TR、AR 与 profit
- ☐ 能区分 economies 与 diseconomies of scale
- ☐ 能比较 labour-intensive 与 capital-intensive methods
- ☐ 能评价 competitive market 与 monopoly 对价格、产量和消费者的影响

单元 4 | 政府与宏观经济 | Government and the macroeconomy

官方范围 | Official scope: 4.1–4.7

1. 单元导读 | Unit introduction

本单元研究政府希望整个经济实现什么，以及如何使用政策工具。核心不是背“政策使经济变好”，而是写出工具变化怎样影响消费、投资、成本、生产率、总需求和实际结果。目标之间可能冲突，因此评价必须说明经济所处阶段、时间滞后、政策幅度和利益相关者。

与人教版衔接及口径差异 | Connection with Chinese curricula. 与人教版宏观调控衔接。Cambridge 把 fiscal policy、monetary policy、supply-side policy 分开考查，并要求计算预算余额、失业率或 CPI 变化。稳定物价不是“所有价格不变”，而是避免过高且不稳定的通胀或持续通缩。

学习路线 | Learning route.

1. 4.1 宏观目标、指标与目标冲突 – Macroeconomic aims and conflicts
2. 4.5–4.7 经济增长、失业、通胀与通缩 – Growth, unemployment, inflation and deflation
3. 4.2–4.4 财政、货币与供给侧政策 – Fiscal, monetary and supply-side policies
4. 4.2–4.7 政策传导、适用条件与评价 – Policy transmission and evaluation

2. 知识点精讲 | Detailed teaching

4.1 | 宏观目标、指标与目标冲突 | Macroeconomic aims and conflicts

常见目标是经济增长、低失业、低且稳定通胀、国际收支平衡与更公平的收入分配。它们会冲突：刺激总需求或许降低周期性失业，但也可能提高通胀和进口，恶化经常账户。

知识点 01/02 | 宏观目标与指标：五类目标与测量口径 | Five macroeconomic objectives and their indicators

主要目标包括 economic growth、low unemployment、price stability、redistribution of income 和 balance of payments stability。Growth 用 real GDP 的变化衡量；通胀用 consumer price index 的持续增长；失业率=unemployed/labour force × 100；收入分配可结合 Lorenz curve；国际交易重点看 current account。

知识点 02/02 | 宏观目标与指标：目标之间既有协同也有冲突 | Synergies and conflicts between policy objectives

衰退时提高 AD 可能同时增加 output 和 employment；接近满负荷时继续刺激 AD 更可能推高 inflation。快速增长可能增加 imports、恶化 current account，也可能通过生产率和出口改善它。重新分配收入可能提高低收入者消费，却可能影响工作、储蓄和投资激励。

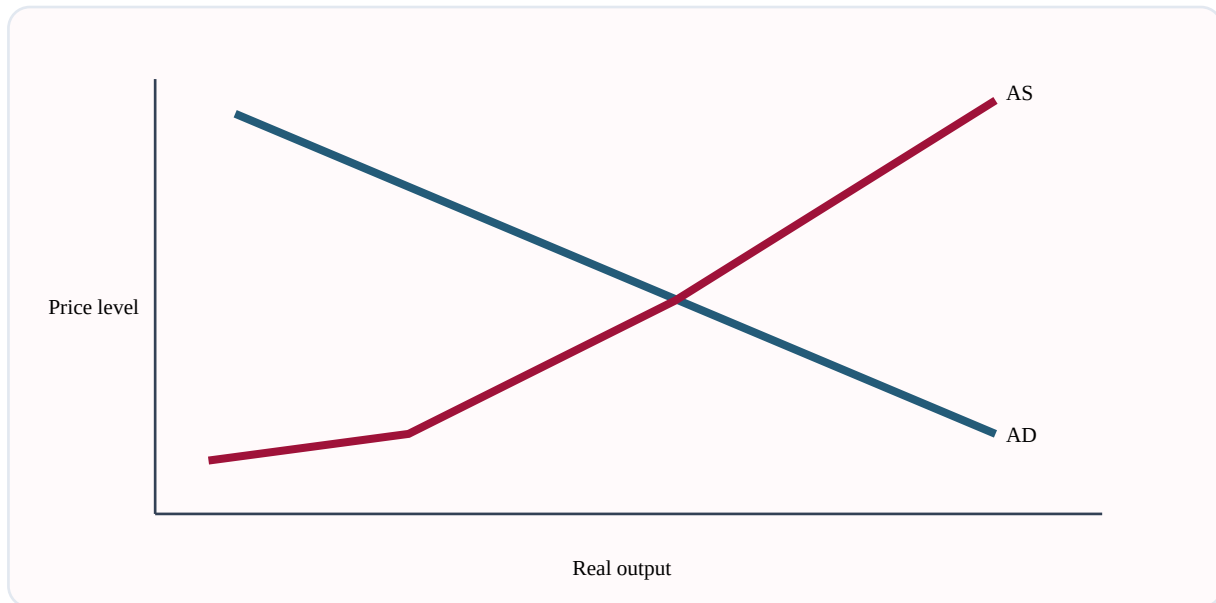
Government 常关注 growth、employment、price stability、current-account stability 与 income distribution。每个目标需要不同指标，指标改善也可能与另一目标冲突。

AD rises when spare capacity exists → firms increase output → derived demand for labour rises → unemployment falls → household income and tax revenue rise；接近满产时则 inflation pressure 更大。

失业人数0.6m、labour force 12m，unemployment rate=5%。若失业人数不变但更多人进入 labour force，失业率可能变化，因此分母也要检查。

Macroeconomic objectives are related but not identical. Economic growth is measured using real output; price stability refers to low and predictable inflation; full employment does not mean zero unemployment because frictional unemployment normally exists.

比较国家时须使用相同年份、定义与单位，并注意 informal sector、underemployment 和地区差异。



图解：总需求增加通常同时提高价格水平和实际产出，但两者幅度取决于闲置资源与供给反应。

4.5–4.7 | 经济增长、失业、通胀与通缩 | Growth, unemployment, inflation and deflation

Economic growth 是实际 GDP 增加；它可提高收入与税收，但若只依赖资源开采，可能污染并加剧不平等。失业分为 frictional、structural、cyclical 和 seasonal。通胀是总体价格水平持续上升，不是单一商品涨价；demand-pull inflation 来自总需求增长过快，cost-push inflation 来自工资、能源或进口成本上升。

知识点 01/02 | 经济增长、失业与通胀：三个概念必须分开 | Distinguishing growth, unemployment and inflation

Economic growth 是 real GDP 增加；recession 通常是 real output 下降。Inflation 是总体价格水平持续上升，不是一个商品涨价。Unemployment 指愿意且能够工作、正在寻找工作却没有工作的人。Frictional、structural、cyclical、seasonal unemployment 的原因和政策不同。

知识点 02/02 | 经济增长、失业与通胀：原因与影响的双向链条 | Causes and consequences in both directions

Demand-pull inflation 来自 AD 增长过快；cost-push inflation 来自工资、能源或进口成本上升。通胀降低固定收入者购买力并增加不确定性，但温和通胀也可能伴随增长。失业降低 household income、tax revenue 和 output，却减少工资与价格压力。增长可来自资源增加、productivity、investment 和技术进步。

Growth 用 real GDP 变化衡量；inflation 用 general price level 的持续变化衡量；unemployment 只包括 willing and able to work 且正在找工作的人。Nominal、real、rate 与 level 必须区分。

oil price rises → firms' production costs rise → SRAS falls → price level rises and real output falls → cost-push inflation and cyclical unemployment may coexist.

CPI 从120升到126， $\text{inflation} = (126 - 120) / 120 \times 100 = 5\%$ 。下一年 CPI 从126升到129.15，通胀2.5%；价格仍上涨，只是上涨速度减慢。

Disinflation is a fall in the inflation rate while the price level continues to rise. Deflation is a sustained fall in the general price level. Structural unemployment reflects a mismatch of skills or location, while cyclical unemployment results from deficient aggregate demand.

Real GDP per head may fall even when real GDP rises if population grows faster.

适度需求增长可能同时降失业并带来温和通胀；长期结果取决于 productive capacity、expectations 和政策可信度。

4.2-4.4 | 财政、货币与供给侧政策 | Fiscal, monetary and supply-side policies

Fiscal policy: 政府增加支出或减税以提高 AD；代价可能是预算赤字与债务。Monetary policy: 降低利率使借贷更便宜，往往提高消费和投资，但在信心低迷时效果弱。Supply-side policy: 培训、教育、基础设施与竞争政策提高生产能力，见效较慢但更能改善长期增长。

知识点 01/02 | 政策工具：三类政策作用点不同 | Where fiscal, monetary and supply-side policies act

Fiscal policy 改变 government spending 与 taxation；monetary policy 通过 interest rates、money/credit conditions 和 exchange rate 影响需求；supply-side policy 改善 education、infrastructure、competition、labour mobility 和 incentives，提高 productive capacity。Expansionary 政策通常提高 AD，contractionary 政策降低 AD。

知识点 02/02 | 政策工具：政策传导不是一句“促进经济” | Developing the policy transmission mechanism

降息 → borrowing cheaper/saving less attractive → C 与 I 可能增加 → AD rises → firms raise output → employment rises。所得税下降 → disposable income rises → consumption rises。培训 → skills/productivity rise → unit cost falls → AS/productive potential rises。每条链都要说明可能条件：confidence、time lag、size、elasticity、debt 和 import leakage。

Fiscal policy 改变 government spending 与 taxation；monetary policy 改变 interest rates 或 credit；supply-side policy 改变资源数量、质量、流动性和生产率。工具必须匹配问题原因。

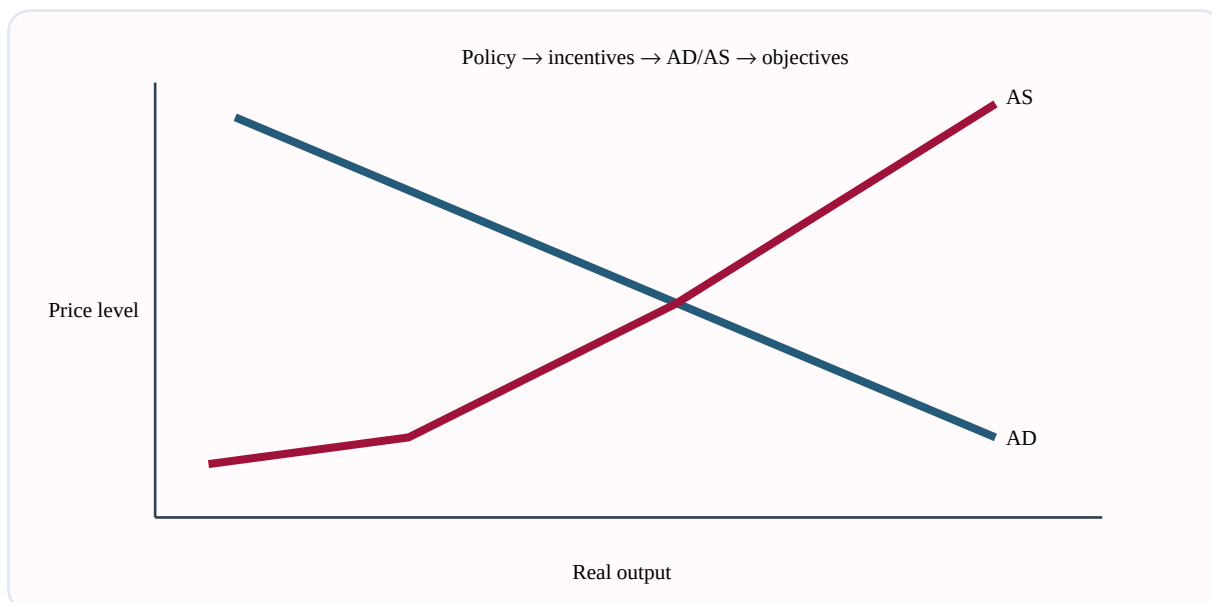
income tax cut → disposable income rises → consumption may rise → AD rises → firms increase output and employment → inflation/imports may also rise。training → skills/productivity rise → productive capacity increases, but after a time lag。

Fiscal policy changes government spending and taxation. Monetary policy changes interest rates or credit conditions. Supply-side policy aims to improve the quantity, quality or mobility of productive resources. Match the policy to the cause of the problem.

Supply-side policy 不一定表示政府减少供给；它旨在提高 economy's productive potential。

评价至少加入 size、timing、confidence、spare capacity、opportunity cost 与 distribution。

Structural unemployment 来自技能或地区错配，retraining 比单纯降息更直接。降息主要提高 AD；若空缺岗位要求的技能不同，需求扩大也未必让失业者立即就业。



图解：政策工具先改变 household/firm incentives，再影响 C、I、cost 或 productivity，最后才影响宏观目标。

4.2–4.7 | 政策传导、适用条件与评价 | Policy transmission and evaluation

不要只写“降低利率促进经济”。拆开写：interest rate falls → monthly loan repayments fall / borrowing becomes cheaper → households consume more and firms invest more → AD rises → output and employment may rise. 然后评价：如果家庭担心失业，可能仍储蓄；若经济接近满产，效果更可能是通胀而不是实际产出。

财政政策同理：government spending on roads → firms receive orders → workers' incomes rise → further consumption occurs. 但政府融资可能增加债务；道路项目也有长期 supply-side benefit。把短期 AD 效果与长期 LRAS 效果分开，答案会更成熟。

知识点 01/02 | 政策传导链的写法：每个箭头都要有 because | Explaining every link in a causal chain

高分 analysis 不是列出变量，而是解释关系。例如 “interest rate ↓ → investment ↑ → AD ↑ → GDP ↑” 仍不够完整；应写借款成本下降提高预期净收益，使更多投资项目有利可图，capital spending 增加，AD 上升，企业面对更高销售而增加产量与就业。

知识点 02/02 | 政策传导链的写法：在链条末端加入限制条件 | Adding a relevant condition to the final outcome

若企业缺乏 confidence，即使利率下降也可能不借款；若经济接近 full employment，AD 增长更多转化为 inflation；若进口倾向高，本国 output 增幅较小。Supply-side policy 需要教育和基建完成后才见效，time lag 往往更长。

政策工具先改变 price、income、cost 或 incentive，随后 households/firms 改变 spending、saving、investment 或 production，最终才影响 AD、output、employment 与 inflation。

interest rate falls → monthly loan cost falls → households may borrow for durable goods → consumption rises → AD rises → firms receive more orders → output and employment rise, if spare capacity exists.

不要用没有题干依据的精确数字，但可以用方向检查。若税后收入由\$1 000升到\$1 100而家庭只消费新增\$60，剩余\$40 saving 是 leakage，AD 增量小于收入增量。

A causal chain should not jump from policy to the final objective. Explain the transmission through incentives and behaviour. Separate a direct effect from secondary effects and avoid repeating the same link in different words.

2027–2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 环境可持续发展作为宏观目标 | Environmental sustainability

2027–2029 考纲明确把 environmental sustainability

列为政府宏观目标。增长若依赖化石燃料、森林砍伐和不可再生资源，短期 real GDP

可能上升，但污染、健康成本和资源耗竭会损害长期生活水平。政府可用 environmental tax、regulation、subsidy for clean technology 和公共交通投资改变激励。

English explanation. Environmental sustainability means meeting present economic needs without seriously reducing the ability of future generations to meet theirs. Faster growth may raise employment and tax revenue, but can also increase emissions and resource depletion. The conflict is not inevitable if cleaner technology and regulation reduce the environmental cost per unit of output.

补充知识点 02/02 | 预算、税收与政策选择 | The government budget and taxation

budget balance = government revenue – government expenditure。结果为正是 surplus，为负是 deficit。税收用于筹资、再分配收入、抑制有害消费、纠正外部成本和管理总需求。评价某种税不能只写“政府收入增加”，还要考虑税基变化、逃税、税负归宿、收入分配和 opportunity cost。

English explanation. A budget deficit occurs when government expenditure exceeds revenue; a surplus occurs when revenue exceeds expenditure. Direct taxes are imposed on income or wealth, while indirect taxes are imposed on spending. Progressive taxation can reduce inequality, but very high marginal rates may weaken incentives or encourage avoidance.

3. 核心词汇表 | Core vocabulary (共 27 词)

English term	中文准确口径	Exam-ready meaning
economic growth	经济增长	An increase in real output over time.
full employment	充分就业	A situation in which unemployment is low apart from unavoidable unemployment.
price stability	物价稳定	A low and stable rate of inflation.
redistribution of income	收入再分配	Government action to change the distribution of income.
balance of payments stability	国际收支稳定	A sustainable relationship between international receipts and payments.
real GDP	实际国内生产总值	Output valued after removing the effect of price changes.
conflict of aims	目标冲突	A situation where pursuing one objective makes another harder to achieve.
recession	经济衰退	A period of falling real output.
unemployment rate	失业率	Unemployed people as a percentage of the labour force.
structural unemployment	结构性失业	Unemployment caused by a mismatch of skills or location.
cyclical unemployment	周期性失业	Unemployment caused by a fall in aggregate demand.
inflation	通货膨胀	A sustained rise in the general price level.
consumer prices index (CPI)	消费者价格指数	An index measuring changes in the price of a representative basket.
fiscal policy	财政政策	Changes in government spending and taxation to influence the economy.
monetary policy	货币政策	Changes in interest rates or money conditions to influence the economy.
supply-side policy	供给侧政策	Measures designed to increase productive capacity and efficiency.

English term	中文准确口径	Exam-ready meaning
budget deficit	财政赤字	Government spending exceeding government revenue.
aggregate demand	总需求	Total planned spending on domestic output.
incentive	激励	A factor encouraging a change in economic behaviour.
time lag	时滞	The delay before a policy has its full effect.
transmission mechanism	传导机制	The stages through which a policy affects economic outcomes.
aggregate supply	总供给	The total output firms are willing and able to produce.
spare capacity	闲置产能	Unused resources that permit output to expand.
business confidence	企业信心	Firms' expectations about future economic conditions.
multiplier effect	乘数效应	An initial spending change causing a larger final change in national income.
environmental sustainability	环境可持续发展	Meeting present needs without seriously limiting future generations.
budget surplus	预算盈余	The amount by which government revenue exceeds expenditure.

4. 单元练习 | Unit practice (共 14 题)

练习 01/14

Question. Which policy outcome shows a conflict: lower unemployment accompanied by higher demand-pull inflation, or higher productivity with lower costs?

练习 02/14

Question. Discuss whether low unemployment is always beneficial to an economy.

练习 03/14

Question. The inflation rate falls from 8% to 3%. Which statement is correct?

练习 04/14

Question. Analyse two costs of high inflation for firms.

练习 05/14

Question. Which policy is most directly designed to reduce structural unemployment: lower interest rates, retraining, higher government consumption, or a currency depreciation?

练习 06/14

Question. Discuss whether higher government spending will increase economic growth.

练习 07/14

Question. Which is the missing link: lower interest rate → ---- → higher investment → higher AD?

练习 08/14

Question. Analyse how a depreciation may reduce unemployment.

练习 09/14

Question. Which policy is most directly supply-side?

A raising government pensions B reducing the policy interest rate C subsidising worker training D increasing an indirect tax to reduce demand

练习 10/14

Question. Government revenue is \$84bn and expenditure is \$97bn. Calculate the budget balance and identify its type. [3]

练习 11/14

Question. A country has a labour force of 25 million and 1.5 million unemployed people. Calculate its unemployment rate. [3]

练习 12/14

Question. Explain how a higher interest rate may reduce demand-pull inflation. [6]

练习 13/14

Question. Analyse one conflict between economic growth and environmental sustainability. [6]

练习 14/14

Question. Discuss whether expansionary fiscal policy is the best way to reduce unemployment. [8]

5. 练习答案与解析 | Answers and explanations (共 14 题)

答案 01/14

正确答案 | Correct answer. Lower unemployment with higher demand-pull inflation, because improvement in one objective is associated with deterioration in another.

答案 02/14

参考答案 | Model answer. It raises output, income, tax revenue and may reduce benefit spending and poverty. However, severe labour shortages can increase wage costs and inflation, reduce competitiveness and make recruitment difficult. Some job movement is useful. The ideal rate depends on productivity and labour mobility.

答案 03/14

正确答案 | Correct answer. The general price level is still rising, but more slowly. This is disinflation, not necessarily deflation.

答案 04/14

参考答案 | Model answer. Uncertain future costs and prices make investment planning harder. If domestic inflation exceeds competitors', exports become less price competitive and imports more attractive. Firms may also face menu costs and pressure for higher wages.

答案 05/14

正确答案 | Correct answer. Retraining, because it reduces the mismatch between workers' skills and available jobs.

答案 06/14

参考答案 | Model answer. Spending raises AD and may increase output when spare capacity exists. Infrastructure and education can also increase long-run productivity. However, spending may cause inflation, imports, taxation or debt, and inefficient projects have opportunity cost. The result depends on spending type and economic conditions.

答案 07/14

正确答案 | Correct answer. Borrowing becomes cheaper or the expected net return on investment rises.

答案 08/14

参考答案 | Model answer. Exports become cheaper in foreign currency and imports become dearer domestically. If demand responds, export sales rise and spending switches to domestic products. Firms increase output, raising derived demand for labour and reducing cyclical unemployment.

答案 09/14

正确答案 | Correct answer. C – Training raises skills, occupational mobility and labour productivity, increasing productive capacity. The other policies mainly affect aggregate demand or prices.

答案 10/14

正确答案 | Correct answer. Budget balance = revenue – expenditure = 84 – 97 = –\$13bn. It is a budget deficit of \$13bn because expenditure exceeds revenue.

答案 11/14

正确答案 | Correct answer. Unemployment rate = unemployed/labour force $\times 100 = 1.5/25 \times 100 = 6\%$.

答案 12/14

参考答案 | Model answer. Higher interest makes borrowing more expensive and saving more attractive. Household consumption may fall, while firms may reduce investment because finance costs rise. Aggregate demand grows more slowly or falls, reducing excess demand and upward pressure on the general price level. The effect depends on debt levels and confidence.

答案 13/14

参考答案 | Model answer. Faster growth may increase production, transport and energy use. If energy relies on fossil fuels, emissions and pollution rise, creating health costs and resource depletion. This can damage future productive capacity and welfare. The conflict is smaller if growth comes from cleaner technology, renewable energy and more efficient resource use.

答案 14/14

参考答案 | Model answer. Higher government spending or lower tax can raise aggregate demand, increasing firms' sales and derived demand for labour, so it is useful for cyclical unemployment when spare capacity exists. It may be less effective for structural unemployment caused by skill or location mismatch. It can create a budget deficit, inflation or imports, and has time lags. A policy mix including training, mobility support and targeted infrastructure is more suitable when unemployment is structural.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能定义并区分六项宏观目标
- ☐ 能解释增长、就业、物价、国际收支、分配和环境目标之间的冲突
- ☐ 能计算 budget deficit/surplus 并分类 direct/indirect、progressive/proportional/regressive tax
- ☐ 能解释 fiscal policy 的完整传导
- ☐ 能解释利率、money supply 和 exchange rate 的货币政策渠道
- ☐ 能比较教育培训、减税、私有化、基础设施等供给侧措施
- ☐ 能区分实际增长与衰退及其原因后果
- ☐ 能计算失业率并识别 frictional/structural/cyclical/seasonal unemployment
- ☐ 能解释 CPI、demand-pull/cost-push inflation 和 deflation
- ☐ 能按时滞、幅度、信心、资源闲置和副作用评价政策

单元 5 | 经济发展 | Economic development

官方范围 | Official scope: 5.1–5.4

1. 单元导读 | Unit introduction

本单元回答“经济变大是否等于生活变好”。学生需要把 real GDP per head、HDI、贫困、人口结构、产业结构、教育健康和收入分配结合起来。任何单一指标都有盲点；高分比较必须同时说明数据所反映的维度、未反映的维度和国情差异。

与人教版衔接及口径差异 | Connection with Chinese curricula.

与人教版共同富裕、人口和可持续发展衔接。Cambridge 常用 real GDP per head 与 HDI 比较国家，同时要求区分 absolute poverty 与 relative poverty，并从 productivity、saving、investment、education、health、natural resources 和 sectoral structure 解释发展差异。

学习路线 | Learning route.

1. 5.1 经济增长与经济发展 – Economic growth and development
2. 5.1 生活水平: real GDP per head 与 HDI – Living standards and development indicators
3. 5.2–5.4 贫困、人口与发展差异 – Poverty, population and development differences
4. 5.2–5.4 发展政策及其评价 – Development policies
5. 5.1–5.4 跨国比较指标的使用与局限 – Comparing development indicators

2. 知识点精讲 | Detailed teaching

5.1 | 经济增长与经济发展 | Economic growth and development

增长是实际 GDP 上升；development（发展）是生活质量和选择能力的更广泛改善。一个国家可因石油出口 GDP 增长，却仍有贫困、教育不足、污染和地区差距。因此答案中把“人均实际 GDP”与健康、教育、住房、自由、安全、环境一起写。

知识点 01/02 | 增长不等于发展：两个概念的范围不同 | Different meanings of growth and development

Growth 是 real GDP 增加，development 是生活质量的广泛改善，包括 income、health、education、housing、environment 和 freedom。增长可增加税基和就业，为发展提供资源；但若收益集中在少数人、污染严重或人口增长更快，平均居民未必受益。

知识点 02/02 | 增长不等于发展：因果关系不是自动发生 | Why growth does not automatically create development

资源出口提高 GDP 后，政府若把收入用于 clean water、schools 和 clinics，human capital 与 life expectancy 可提高；若收入外流、腐败或只进入资本密集行业，poverty 可能仍高。Real GDP per head 比总 GDP 更接近平均物质生活，但仍忽略 distribution 和非货币因素。

Growth 观察 real output；development 观察 income、health、education、opportunity、distribution 和 environment。GDP 是生产指标，不是完整 welfare 指标。

export-led growth → employment/income/tax revenue rise → government may fund health and education → human capital and life expectancy improve → development；若利润外流或污染严重，链条可能中断。

Real GDP 增6%、人口增4%，real GDP per head 大致增约2%，不是6%。若最高10%获得大部分新增收入，median household 的生活水平可能几乎不变。

Growth expands the amount an economy produces. Development concerns what the output enables people to achieve. Per-head measures, distribution and non-material indicators are therefore essential when assessing welfare.

结论最好区分 short run material gain 与 long-run sustainable development.

5.1 | 生活水平: real GDP per head 与 HDI | Living standards and development indicators

GDP per head 适合粗略比较物质产出, 但忽略收入分配、非市场生产、工作时长与环境。HDI 结合寿命、教育和收入, 覆盖更广; 仍是平均数, 会掩盖性别、城乡与地区不平等。比较国家时还要注意汇率、购买力和人口结构。

知识点 01/02 | 衡量生活水平: 单一指标与综合指标 | Single indicators and composite measures

Real GDP per head = real GDP / population, 适合比较平均物质产出, 但跨国比较还要考虑 exchange rate、purchasing power、informal economy 和 data accuracy。HDI 结合 income、education 和 life expectancy, 范围更广, 却仍是平均值, 不能完全反映 inequality、security 或 environment。

知识点 02/02 | 衡量生活水平: 读数据要比较、联系并找例外 | Comparing data, identifying relationships and exceptions

一国人均收入高而 life expectancy

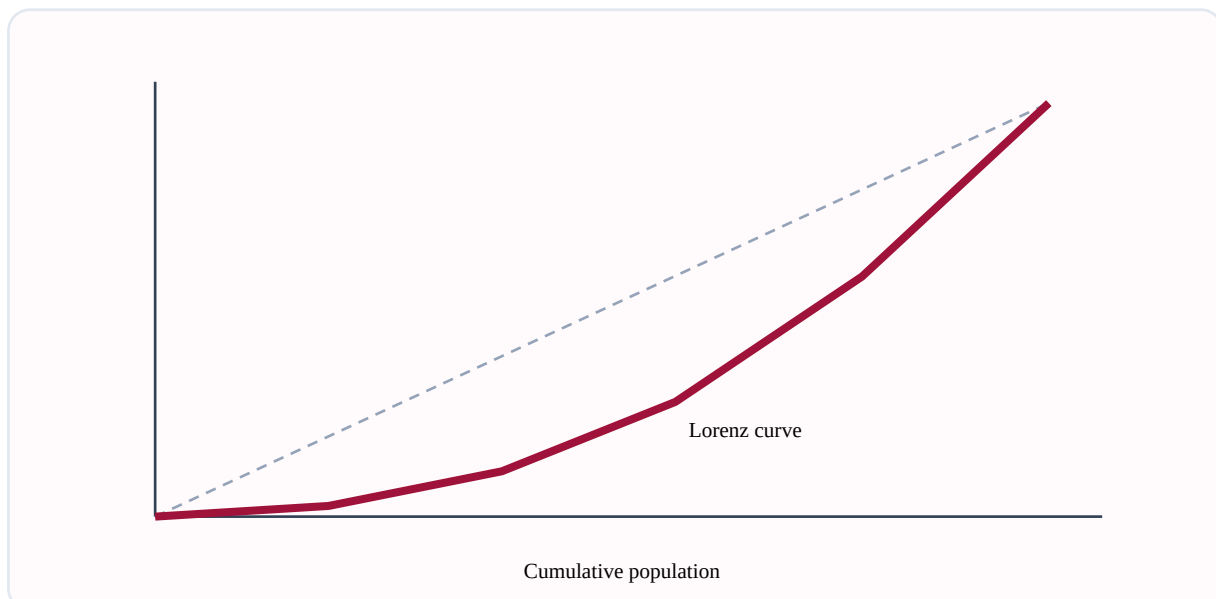
低, 可能说明不平等、医疗不足或污染; 另一国收入较低但教育健康良好, 可能源于公共服务效率。Correlation 不等于 causation, 还要考虑其他变量。比较时应使用同一年、相同单位和 real/per-head 数据。

Real GDP per head 调整 inflation 与 population; HDI 加入 education 与 life expectancy; 单项社会指标可补充 sanitation、nutrition 或 mortality。不同指标回答不同问题。

A国 real GDP 500bn、人口50m, 人均10 000; B国300bn、人口20m, 人均15 000。A总产出更高, 但B平均物质产出更高。还不能据此判断分配和健康。

Real GDP per head is useful for average material output, while HDI includes income, education and life expectancy. Purchasing-power adjustments improve international comparison. No single index captures inequality, freedom, safety or environmental quality.

HDI 较高不表示每个分项都较高, 也不代表国内人人处于相同水平。



图解: Lorenz curve 越远离完全平等线, 收入分配通常越不均; 它描述 distribution, 不直接测量绝对贫困。

5.2–5.4 | 贫困、人口与发展差异 | Poverty, population and development differences

绝对贫困指基本需要难以满足；相对贫困强调与社会正常生活水平的差距。人口快速增长可扩大劳动年龄人口和市场，也可能压低人均资本、增加学校医疗负担与失业。低储蓄、基础设施不足、政治不稳定、初级产品依赖和债务均可形成发展陷阱。

知识点 01/02 | 贫困、人口与发展障碍：贫困的两种口径 | Absolute and relative measures of poverty

Absolute poverty 指收入不足以满足 food、shelter 等基本需要；relative poverty 指收入显著低于社会通常水平。低收入→营养教育不足→productivity 低→工资低，形成 poverty cycle。税收转移、教育医疗和就业增长可能打破循环，但政策有财政成本和激励问题。

知识点 02/02 | 贫困、人口与发展障碍：人口数量之外还有结构 | Population size, structure and dependency

Birth rate、death rate、life expectancy 和 net migration 改变人口规模；age structure 与 dependency ratio 影响 labour supply、tax revenue 和公共支出。年轻人口可能带来未来劳动力优势，也需要当前教育投入；老龄化增加养老金和医疗压力，但也可能扩大养老服务需求。

Absolute poverty 看 basic needs，relative poverty 看相对社会标准；population analysis 还要区分 birth rate、death rate、migration、age structure 与 dependency。

low income → poor nutrition/education → low productivity → low wages → limited saving/investment → poverty persists。政策可在 education、health、credit 或 job creation 处打断循环。

工作年龄人口600万，儿童250万，老人150万，dependency ratio=(250+150)/600×100=66.7%。这不表示每100名工人实际供养66.7人，因为工作年龄者并非全就业。

Absolute poverty concerns inability to meet basic needs. Relative poverty compares a household with prevailing living standards. A high dependency ratio may increase pressure on workers and public services, but dependants can also contribute unpaid work and future labour.

低 birth rate 不必然使人口立即下降；migration、death rate 和 population momentum 也会影响。

人口结果取决于年龄结构、人力资本、女性就业、工作机会和基础设施，而非只看人数。

5.2-5.4 | 发展政策及其评价 | Development policies

教育、医疗、基础设施、微型信贷、贸易开放、FDI 与援助都可能有用。例：修港口可降低出口运输成本，但若政府腐败、维护资金不足，项目效益会很低。评价要问：谁受益？资金来自哪里？短期与长期效果？环境是否可持续？

知识点 01/02 | 发展策略的评价：政策通过生产率改变发展 | How policy raises productivity and development

Education raises skills and occupational mobility；healthcare reduces illness and improves effective labour supply；infrastructure lowers transport and communication costs；microfinance may support enterprise；property rights and stable institutions encourage investment。短期补贴能缓解贫困，但长期发展通常要求生产能力与机会增加。

知识点 02/02 | 发展策略的评价：评价预算、时滞与受益者 | Budgets, time lags and the distribution of benefits

同一笔预算投向高速公路会产生 opportunity cost，例如少建学校。大型项目可能偏向城市，农村贫困者未必直接受益。教育见效慢但外部收益广；现金转移见效快却需持续财政来源。Foreign aid 和 FDI 可带来资金技术，也可能有条件、利润外流和 dependency。

Education、healthcare、infrastructure、redistribution、aid 与 FDI 针对的 binding constraint 不同。比较政策要看 mechanism、time lag、cost、coverage 和 sustainability。

rural road investment → transport time/cost falls → farmers access larger markets → spoilage falls and revenue rises → saving/investment and rural living standards may improve。

预算\$50m可建学校或发一年补贴。补贴立刻增加可支配收入；学校需多年才转化为技能，但收益可能持续更久。两者的 opportunity cost 与时间结构不同。

A development strategy should address the binding constraint. Education is weak if graduates have no jobs; roads are less useful without maintenance; foreign investment brings fewer benefits when local firms and workers cannot form linkages.

投入资金不等于 outcome 改善；必须说明服务质量、覆盖人群与执行能力。

“最佳”要求比较替代政策，并结合该国最主要障碍作判断。

5.1-5.4 | 跨国比较指标的使用与局限 | Comparing development indicators

看到“哪个国家生活水平更高”，不要只比较一个 GDP 数字。依次问：是否是 real GDP？是否 per head？汇率是否反映购买力？收入是否平均分配？教育、寿命、饮水和环境怎样？一个资源出口国的人均 GDP 高，仍可能有很大贫富差距。

HDI 的读法：它把 income、education 与 life expectancy 组合，所以能补充 GDP；但它是国家平均值，无法显示一个城市很富而农村很穷。可直接写：HDI is broader than GDP per head, although both figures can hide inequality within a country.

知识点 01/02 | 指标题的比较框架：指标比较的四层结构 | A four-stage method for comparing indicators

第一层确认定义与单位；第二层描述 trend/difference；第三层用数据验证 relationship；第四层指出 exception 与 limitation。若比较 GDP，先确认 nominal or real、total or per head。若比较失业率，关注 labour-force 定义是否一致。

知识点 02/02 | 指标题的比较框架：平均值不代表每个人 | Why an average does not represent every household

GDP per head 上升可能与 median income 不同方向；HDI 改善也可能隐藏地区差异。数据只显示 correlation 时，不能直接写 A causes B。需要提出合理机制，并承认 reverse causation 或 third variable。

Describe 只陈述数据，compare 必须涉及双方，calculate 展示公式，analyse 把数据证据与经济机制连接。题目动词决定答案任务。

失业率从4%到6%是上升2 percentage points，也是相对上升50%。题目若未指定，要用最清楚的表述，不能含混写“上升2%”。

Use evidence selectively. A strong relationship answer states the overall pattern, cites two relevant comparisons, identifies an exception and offers an economic explanation. Correlation alone does not prove causation.

不要只抄最高值与最低值；relationship 题需要跨多组数据，exception 也必须有数值证据。

2027-2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 环境与贫困的双向关系 | Environment and poverty

环境恶化既可能是贫困原因，也是贫困结果。干旱、洪水和土壤退化降低农业产量与收入；低收入家庭又可能依赖砍柴、过度放牧或污染性燃料来满足短期需要。政策应把灌溉、清洁能源、产权、保险、教育和替代就业结合起来，而不是把责任简单归于个人选择。

English explanation. Environmental damage can create a poverty trap. Lower crop yields reduce income and saving, which limits investment in resilient technology; future productivity then remains low. Policies are more effective when environmental protection is combined with access to finance, education, infrastructure and alternative livelihoods.

补充知识点 02/02 | 人口结构与最佳人口 | Population structure and optimum population

人口增加既扩大 labour force 和 market size, 也增加教育、医疗、住房和就业压力。高 dependency ratio 表示相对较少的劳动年龄人口负担较多儿童或老人。optimum population 不是固定人数, 而是在现有资源、资本和技术条件下使 output per head 最大的人口规模。

English explanation. Population growth can raise total output without raising output per head. The effect depends on age structure, skills, employment opportunities, capital and natural resources. An ageing population may increase pension and healthcare spending, while a young population requires education now but can expand the future labour force.

3. 核心词汇表 | Core vocabulary (共 35 词)

English term	中文准确口径	Exam-ready meaning
economic growth	经济增长	An increase in real GDP.
economic development	经济发展	An improvement in living standards and economic wellbeing.
real GDP per head	人均实际GDP	Inflation-adjusted output divided by population.
living standards	生活水平	Material and wider conditions affecting people's wellbeing.
income distribution	收入分配	How national income is shared among people or households.
sustainability	可持续性	Meeting present needs without damaging future productive or environmental capacity.
GDP per head	人均GDP	GDP divided by population.
Human Development Index (HDI)	人类发展指数	A composite measure using income, education and life expectancy.
life expectancy	预期寿命	The average number of years a person is expected to live.
literacy rate	识字率	The percentage of people able to read and write.
infant mortality rate	婴儿死亡率	Deaths of infants relative to live births.
composite indicator	综合指标	A measure combining several separate indicators.
informal economy	非正规经济	Economic activity not fully recorded in official statistics.
absolute poverty	绝对贫困	Insufficient income to meet basic needs.
relative poverty	相对贫困	Income far below the normal standard in a society.
dependency ratio	抚养比	Dependants relative to the working-age population.
birth rate	出生率	Live births per thousand people per year.
death rate	死亡率	Deaths per thousand people per year.
net migration	净迁移	Immigration minus emigration.
poverty cycle	贫困循环	A self-reinforcing process linking low income, low investment and low productivity.
human capital	人力资本	Workers' knowledge, skills and health.
infrastructure	基础设施	Transport, energy, communications and other systems supporting production.
foreign direct investment (FDI)	外国直接投资	Investment giving a foreign firm lasting control of production.
foreign aid	外国援助	Resources transferred to a country on concessional terms.

English term	中文准确口径	Exam-ready meaning
redistribution	再分配	Policies that alter the distribution of income or wealth.
microfinance	小额信贷	Small-scale financial services for low-income borrowers.
diversification	多元化	Reducing dependence by developing a wider range of activities.
percentage-point change	百分点变化	The arithmetic difference between two percentage rates.
percentage change	百分比变化	The change divided by the original value, multiplied by 100.
index number	指数	A number showing a value relative to a base period.
correlation	相关性	A statistical relationship between two variables.
causation	因果关系	A relationship in which one change produces another.
real value	实际值	A value adjusted for changes in the price level.
per-head measure	人均指标	A total divided by the relevant population.
optimum population	最佳人口	The population size that maximises output per head under existing conditions.

4. 单元练习 | Unit practice (共 19 题)

练习 01/19

Question. Which most clearly indicates development rather than growth alone: higher real GDP, longer life expectancy, higher price level, or more currency in circulation?

练习 02/19

Question. Discuss whether rapid economic growth improves living standards.

练习 03/19

Question. Why is real GDP per head usually better than nominal GDP for comparing living standards over time?

练习 04/19

Question. Analyse two limitations of using GDP per head to compare countries.

练习 05/19

Question. Country A has real GDP \$240bn and population 60m. Country B has real GDP \$180bn and population 30m. Calculate real GDP per head and compare material living standards. [4]

练习 06/19

Question. Which change directly increases an ageing dependency ratio: higher birth rate, longer life expectancy with low fertility, net immigration of young workers, or later retirement?

练习 07/19

Question. Discuss whether population growth benefits a developing economy.

练习 08/19

Question. A country has 18m people aged 0–14, 54m aged 15–64 and 12m aged 65+. Calculate the age dependency ratio. [3]

练习 09/19

Question. Which policy most directly raises human capital: a tariff, teacher training, currency appreciation, or a maximum price?

练习 10/19

Question. Discuss whether foreign aid is the best way to promote development.

练习 11/19

Question. Country X has twice the GDP of Y but four times the population. Which has higher GDP per head?

练习 12/19

Question. Analyse a positive relationship between education spending and labour productivity shown in a table.

练习 13/19

Question. Unemployment rises from 4% to 6%. State the change in percentage points and the percentage increase. [3]

练习 14/19

Question. Which item is included in HDI?

A inflation rate B life expectancy C current-account balance D income inequality

练习 15/19

Question. Real GDP rises from \$480bn to \$504bn while population rises from 40m to 42m. Calculate real GDP per head before and after. Has it increased? [4]

练习 16/19

Question. Explain why a high real GDP per head may coexist with poverty. [4]

练习 17/19

Question. Analyse how improved healthcare may promote economic development. [6]

练习 18/19

Question. Explain two economic effects of an ageing population. [6]

练习 19/19

Question. Discuss whether rapid population growth will reduce living standards in a developing economy. [8]

5. 练习答案与解析 | Answers and explanations (共 19 题)

答案 01/19

正确答案 | Correct answer. Longer life expectancy, because it directly reflects a health and welfare outcome.

答案 02/19

参考答案 | Model answer. Growth may create jobs, raise incomes and tax revenue for services. It can finance infrastructure. However, population may grow faster, income may be unequal, work hours may rise and pollution or resource depletion may worsen. The result depends on inclusiveness and sustainability.

答案 03/19

正确答案 | Correct answer. It adjusts for both price changes and population, though it remains an average.

答案 04/19

参考答案 | Model answer. Exchange rates may not reflect local purchasing power, so equal dollar incomes buy different amounts. Informal and unpaid activity may be recorded differently. The average also hides inequality, while pollution and leisure are omitted.

答案 05/19

正确答案 | Correct answer. A: $\$240\text{bn} / 60\text{m} = \$4,000$ per head. B: $\$180\text{bn} / 30\text{m} = \$6,000$ per head. B has the higher average real output per person, although this alone does not prove every resident is better off because distribution and non-material factors are omitted.

答案 06/19

正确答案 | Correct answer. Longer life expectancy with low fertility, because the elderly share rises relative to the working-age population.

答案 07/19

参考答案 | Model answer. It can expand labour supply, demand and tax base, encouraging investment. A young population can become a demographic dividend. But education, housing and jobs must expand; otherwise unemployment, congestion and pressure on resources may rise.

答案 08/19

正确答案 | Correct answer. Dependent-age population = $18\text{m} + 12\text{m} = 30\text{m}$. Dependency ratio = $30/54 \times 100 = 55.6\%$. This is an age-structure measure, not the ratio of non-workers to actual workers.

答案 09/19

正确答案 | Correct answer. Teacher training, because it can improve the quality of education and workers' future skills.

答案 10/19

参考答案 | Model answer. Aid can finance health, education, disaster relief and infrastructure where domestic revenue is low. It may transfer expertise. However, tied aid, corruption, dependency and poor project selection reduce benefits. Trade access, governance or domestic taxation may be more sustainable.

答案 11/19

正确答案 | Correct answer. Country Y. X has only half Y's GDP per head when both totals are divided by population.

答案 12/19

参考答案 | Model answer. Higher spending may improve teacher quality and access, raising workers' skills. Skilled workers use capital more effectively and produce more per hour. Cite supporting countries and an exception; productivity also depends on technology, health and spending efficiency.

答案 13/19

正确答案 | Correct answer. The rate rises by 2 percentage points. Relative to the original rate, the percentage increase is $(6 - 4)/4 \times 100 = 50\%$. The two figures answer different questions.

答案 14/19

正确答案 | Correct answer. B – HDI combines a health indicator, education indicators and income per head. It does not directly include inflation, the current account or income inequality.

答案 15/19

正确答案 | Correct answer. Before: $\$480\text{bn}/40\text{m} = \$12\,000$. After: $\$504\text{bn}/42\text{m} = \$12\,000$. Real GDP rises, but real GDP per head is unchanged, so this measure alone shows no rise in average material living standards.

答案 16/19

参考答案 | Model answer. Real GDP per head is an average and does not show income distribution. A large share of income may go to a small group, leaving others below an absolute or relative poverty line. It also omits unequal access to health, education and public services.

答案 17/19

参考答案 | Model answer. Better healthcare reduces illness and absenteeism, increasing labour productivity and effective labour supply. Longer healthy lives can raise income, saving and tax revenue, allowing more investment in education and infrastructure. However, healthcare spending has an opportunity cost and benefits may take time.

答案 18/19

参考答案 | Model answer. A larger elderly population may raise government pension and healthcare spending, creating pressure for higher taxes or lower spending elsewhere. A smaller working-age share may reduce labour supply and raise wages, although later retirement, migration or automation can offset this.

答案 19/19

参考答案 | Model answer. It may increase dependency, pressure schools, housing and jobs, and reduce capital per worker, lowering GDP per head. It can also expand the future labour force and market, attract investment and permit specialisation. The result depends on education, job creation, infrastructure, natural resources, age structure and whether output grows faster than population.

6. 自学检查清单 | Self-study checklist (共 11 项)

- ☐ 能区分 real GDP growth 与 economic development
- ☐ 能计算或解释 real GDP per head
- ☐ 能说明 HDI 的收入、教育和健康维度
- ☐ 能评价 GDP per head 与 HDI 的优点和局限
- ☐ 能区分 absolute 与 relative poverty
- ☐ 能解释贫困形成的多环节循环
- ☐ 能分析 birth rate、death rate、net migration 与人口变化

- ☐ 能解释 age structure、dependency ratio 和 optimum population
- ☐ 能比较农业、制造业和服务业占比反映的发展阶段
- ☐ 能用 productivity、saving、investment、education、health、resources 解释国家差异
- ☐ 能评价发展政策的机会成本、时滞、执行能力和分配效果

单元 6 | 国际贸易与全球化 | International trade and globalisation

官方范围 | Official scope: 6.1–6.4

1. 单元导读 | Unit introduction

本单元从机会成本出发解释国家为何专业化与贸易，再分析自由贸易、保护措施、跨国公司、汇率和经常账户。比较优势是最容易因“产量更高”而判断错的概念；汇率题则必须先辨明本币升值还是贬值，再写出口进口价格、需求量和经常账户的完整链。

与人教版衔接及口径差异 | Connection with Chinese curricula. 与人教版经济全球化和开放型经济衔接。Cambridge 要求学生用 opportunity cost 判断 comparative advantage, 区分 tariff、quota、subsidy、embargo, 并理解 current account 的 goods、services、primary income、secondary income 四部分。不能把 current account 等同于全部 balance of payments。

学习路线 | Learning route.

1. 6.1 专业化、绝对优势与比较优势 – Specialisation and comparative advantage
2. 6.1 比较优势的数值计算方法 – Numerical comparative advantage
3. 6.1–6.2 自由贸易、贸易保护与全球化 – Free trade, protection and globalisation
4. 6.2 跨国公司与全球生产 – Multinational companies
5. 6.3 汇率、升值与贬值 – Exchange rates, appreciation and depreciation
6. 6.4 经常账户及其变动 – The current account of the balance of payments

2. 知识点精讲 | Detailed teaching

6.1 | 专业化、绝对优势与比较优势 | Specialisation and comparative advantage

Specialisation 让个人、企业或国家集中在机会成本较低的生产活动。关键是 comparative advantage (比较优势)，不是绝对产量。即使 A 国两种产品都生产得更多，只要 A 放弃制造品生产食品的代价较低，它仍应专业化于食品，双方通过贸易可消费到 PPC 以外的组合。

知识点 01/02 | 专业化与比较优势：分工、专业化与交换 | Division of labour, specialisation and exchange

Specialisation 让工人、企业、地区或国家集中于较少任务，可提高 skill、速度和 machinery utilisation；风险是 boredom、dependence 和需求变化。国家贸易方向应由 comparative advantage 决定，即较低 opportunity cost，而非只看 absolute output。

知识点 02/02 | 专业化与比较优势：机会成本数值法 | Calculating comparative opportunity cost

A 每小时可生产 8 cloth 或 4 computers, 因此 1 computer 成本为 2 cloth; B 可生产 6 cloth 或 6 computers, 1 computer 成本为 1 cloth, 所以 B 在 computers 有比较优势。反过来, A 的 1 cloth 成本为 0.5 computer, 低于 B 的 1, 因此 A 在 cloth 有比较优势。互利交换比率必须位于两国机会成本之间。

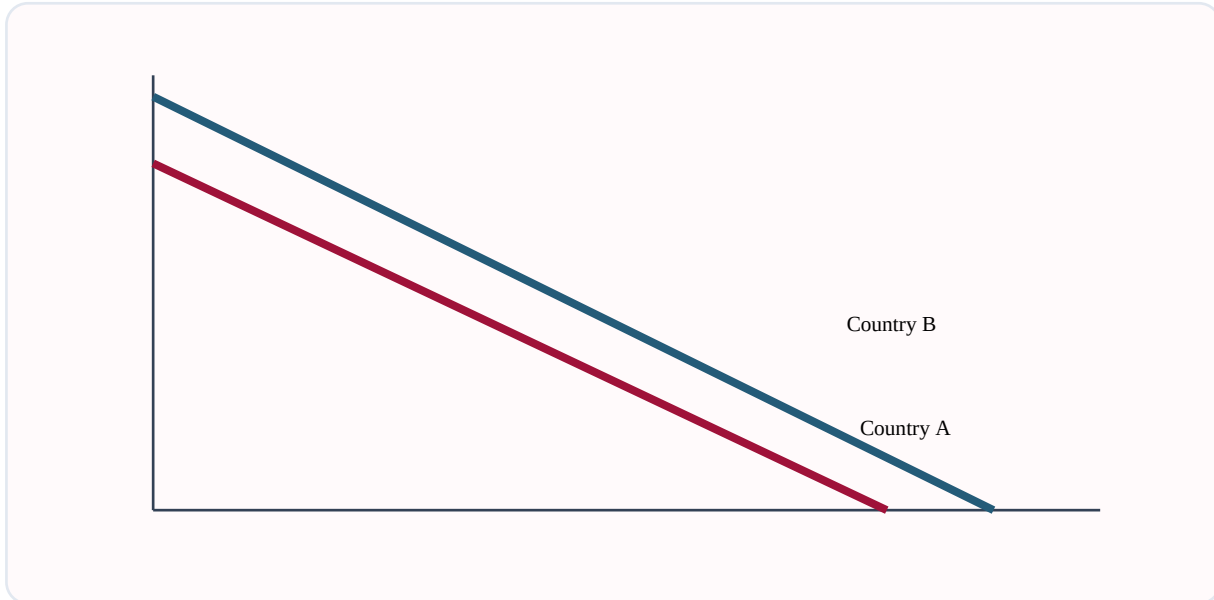
Absolute advantage 比较相同资源下产量, comparative advantage 比较单位产品 opportunity cost。Specialisation 的收益来自 productivity、scale 与 exchange, 代价来自 dependence 和 adjustment。

specialisation → repeated practice and capital designed for a task → productivity rises → unit cost falls → output/trade may increase; 同时 dependence 和 structural unemployment risk 可能上升。

A: 8 cloth或4 computers, 1 computer=2 cloth; B:6 cloth或6 computers, 1 computer=1 cloth。B在computer有比较优势, A在cloth有比较优势。

Comparative advantage explains potential gains from trade because resources are allocated to lower-opportunity-cost production. The model assumes transferable resources, acceptable transport costs and mutually beneficial terms of trade.

不要用“谁产得多”判断 comparative advantage; 必须比较 opportunity cost.



图解: 比较优势取决于较低机会成本。专业化后在双方可接受的交换比率交易, 才可能让两国消费超过各自生产组合。

6.1 | 比较优势的数值计算方法 | Numerical comparative advantage

若 A 国一小时产 8 件衣服或 4 台电脑, 生产 1 台电脑的机会成本是 2 件衣服; B 国一小时产 6 件衣服或 6 台电脑, 生产 1 台电脑的机会成本是 1 件衣服。因此 B 在电脑有比较优势, A 在衣服有比较优势。比较的是比率, 不能只看“谁生产得更多”。

汇率题先说明报价方式。若 1 USD 从 7.0 CNY 升至 7.3

CNY, 人民币对美元贬值; 中国出口对美国买家更便宜, 但进口原材料更贵。紧接着加一句条件: The effect on the current account depends on price elasticities and the time needed for contracts to change.

知识点 01/02 | 比较优势的数值法: 先统一“每一单位”的方向 | Using a consistent per-unit opportunity cost

若表格给的是相同资源下最大产量, 用“另一商品产量 ÷ 本商品产量”求一单位机会成本。A 可产 12 wheat 或 6 cars, 则 1 car=2 wheat, 1 wheat=0.5 car。每个国家都算两行, 再横向比较同一商品; 机会成本较低者有 comparative advantage。

知识点 02/02 | 比较优势的数值法: 检验交换比率是否互利 | Testing whether the terms of trade benefit both countries

Terms of trade 必须介于两国国内机会成本之间。若 A 生产 1 car 放弃 2 wheat, B 放弃 4 wheat, 国际交换率为 1 car:3 wheat, 则 A 出口 car 可换 3 wheat, 超过自行转产得到的 2; B 进口 car 只付 3 wheat, 少于自行生产放弃的 4。

对每个国家分别计算 1 unit X 的 Y opportunity cost 和 1 unit Y 的 X opportunity cost, 再横向比较同一商品。两个单位成本互为倒数。

A:12 wheat或6 cars, 所以1 car=2 wheat; B:8 wheat或2 cars, 所以1 car=4 wheat。A出口car。若1 car换3 wheat, A得到超过2, B支付少于4, 双方获益。

分子分母颠倒会同时错判两个产品。答案必须写单位, 例如 wheat per car。

A 的 1 car=2 wheat, B 的 1 car=4 wheat; 若 terms of trade 为 1 car:3 wheat, A 出口 car 得到 3 wheat, B 进口 car 只付 3 wheat, 双方都优于国内机会成本。

6.1-6.2 | 自由贸易、贸易保护与全球化 | Free trade, protection and globalisation

自由贸易可降低价格、扩大选择、形成规模经济并增强竞争。保护主义包括 tariff、quota、subsidy 和 regulations; 理由可为幼稚产业、就业、国家安全或反倾销, 但代价通常是消费者价格更高、报复与低效率。高分答案必须同时写受保护企业和消费者, 而非只写“有利/不利”。

知识点 01/02 | 自由贸易与保护主义: 自由贸易的收益与调整成本 | Benefits of free trade and adjustment costs

Free trade 扩大 market size, 促进 specialisation、competition、choice 和 economies of scale, 并降低进口价格。然而进口竞争会使缺乏竞争力的企业缩小, 造成 structural unemployment; 国家也可能依赖关键进口或承受外部冲击。

知识点 02/02 | 自由贸易与保护主义: 保护工具与利益相关者 | Protection instruments and stakeholder effects

Tariff 是进口税, 提高进口价格并产生政府收入; quota 直接限制数量, 可能产生配额租; subsidy 降低国内企业成本; embargo 禁止贸易。保护可支持 infant industry、employment、strategic goods 和 current account, 却提高消费者价格、削弱竞争, 并可能引发 retaliation。

Tariff 改变进口价格并产生政府收入, quota 直接限制数量, subsidy 支持国内成本, embargo 禁止交易。每种工具对消费者、生产者与政府的影响不同。

tariff on imported steel $\rightarrow$ import price rises $\rightarrow$ demand shifts towards domestic steel $\rightarrow$ local output/jobs may rise $\rightarrow$ car makers pay more $\rightarrow$ car prices rise and export competitiveness may fall $\rightarrow$ retaliation risk.

世界价\$100, tariff

20%, 进口价约\$120。若进口量从1m降至0.7m, 理论关税收入约\$14m; 但忽略行政成本和可能的逃税。

Protection can support infant industries or strategic capacity, but it reduces competitive pressure and creates costs for users of imported inputs. Quotas restrict quantity directly; tariffs change price and raise government revenue.

Tariff 不会机械地把 import quantity 降为零; 结果取决于需求、国内供给与税率。

结论要比较短期 adjustment benefits 与长期 efficiency costs, 并考虑 retaliation。

6.2 | 跨国公司与全球生产 | Multinational companies

全球化提高资本、信息、商品和人员流动, 可能带来 FDI、就业和技术转移, 也可能令弱劳动者面临跨国竞争、利润外流和文化/环境压力。判断题应结合一个国家的技能、制度与产业结构。

知识点 01/02 | 全球化: 全球化不只是贸易增加 | Globalisation beyond an increase in trade

Globalisation 指商品、服务、资本、技术、信息和生产跨国联系增强。Multinational companies 通过 FDI 建厂, 可能带来 capital、employment、training、tax revenue 和 access to export markets; 东道国也可能面对 profit repatriation、tax avoidance、environmental damage 和对本地企业的竞争。

知识点 02/02 | 全球化: 不同群体获得不同结果 | Why globalisation creates different stakeholder outcomes

消费者可能因进口和竞争获得更低价格与更多选择; 出口企业进入更大市场; 部分工人获得岗位, 另一些工人因 offshoring 失业。来源国企业可降低成本, 却可能减少本国某些就业。评价必须分别讨论 consumers、workers、firms、government 和 environment。

Globalisation 包括 trade、FDI、capital、technology、information 与 labour 的跨国联系。MNC 是参与者之一, 不能把两者当作同义词。

MNC builds factory → capital and jobs enter host economy → workers receive income/training → local suppliers gain orders → tax revenue may fund services; 若 inputs imported and profits repatriated, local multiplier 较小。

工厂销售\$100m, 但进口投入\$45m、利润汇回\$25m, 本地工资税收采购只有\$30m。Gross sales 不能代表留在东道国的全部收益。

Technology transfer and access to global markets can raise productivity. However, bargaining power, tax rules, environmental regulation and local linkages determine how gains are shared. Globalisation creates winners and losers within countries.

短期 job creation 与长期 capability building 要分开, 尤其关注技能转移和本地供应链。

6.3 | 汇率、升值与贬值 | Exchange rates, appreciation and depreciation

汇率是一个货币用另一个货币表示的价格。货币 depreciation (贬值) 使出口在外国买家眼中更便宜、进口在国内更贵; 若需求有弹性, 净出口可能增加。但进口原料更贵也会推高成本型通胀, 而且短期合约会形成 J-curve。不要写“贬值一定改善贸易差额”。

知识点 01/02 | 汇率与经常账户: 汇率是两种货币的价格 | The exchange rate as the price of one currency

Appreciation 表示本币能兑换更多外币, 使 exports 在国外更贵、imports 对本国更便宜; depreciation 相反。浮动汇率受货币 demand 与 supply 影响: export demand、capital inflow 和高利率可能增加本币需求; import spending 与 capital outflow 增加本币供给。

知识点 02/02 | 汇率与经常账户: 数量反应决定最终结果 | Why quantity responses determine the final outcome

贬值首先改变价格, 但 current account 是否改善取决于 PED、替代品、质量、capacity 和 time lag。进口原材料昂贵会造成 cost-push inflation, 抵消出口价格优势。短期合同固定时数量反应慢, 可能出现 J-curve。

Exchange rate 是两种货币的价格。Appreciation/depreciation 先改变 exports 和 imports 的相对价格, 随后才通过 PED、contracts 与 capacity 改变数量和 current account。

currency depreciates → exports cheaper to foreigners/imports dearer domestically → X quantity may rise and M quantity fall → net exports and employment may rise; imported inputs become dearer → costs/inflation may rise。

£ 1=\$1.40变为 £ 1=\$1.20, 英镑

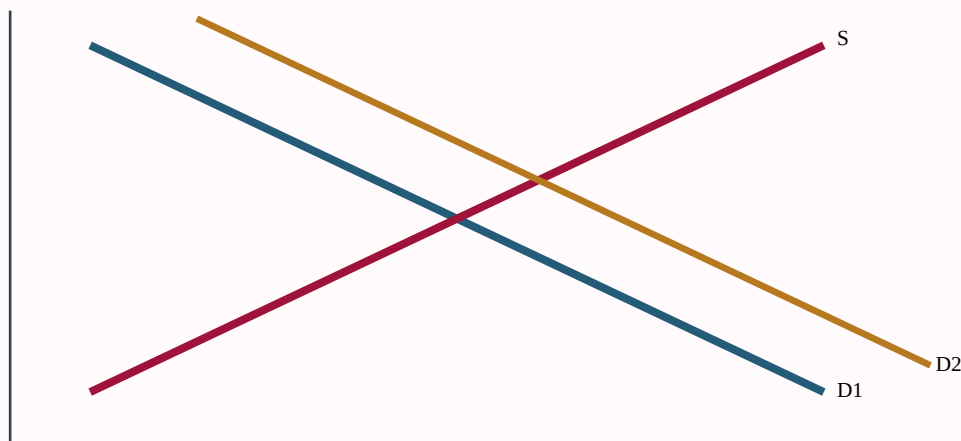
depreciation。售价 £ 100的商品由\$140降至\$120 (若英镑价不变); 英国买\$120商品由约 £ 85.71升至 £ 100。

纵轴标本币价格或 exchange rate, 横轴标 quantity of

currency。需求增加通常使本币升值; 随后分别分析出口价格、进口价格及数量反应, 不能从汇率直接跳到 current-account 结论。

An exchange-rate change first alters relative prices. The current account changes only if buyers respond. Elasticity, contracts, production capacity and imported inputs determine the size and timing of the quantity response.

Appreciation 不是“出口数量必然下降”; 先是相对价格改变, 再由 PED 决定数量。



图解： 本币需求增加可使其升值；升值使 exports dearer、imports cheaper，数量效果取决于弹性。

6.4 | 经常账户及其变动 | The current account of the balance of payments

经常账户记录货物、服务、收入和转移支付。赤字代表流出大于流入，不必然坏：若进口的是资本设备，可能提升未来生产力；但长期靠短期借款融资会有风险。分析时区分“赤字的原因”和“赤字本身”。

知识点 01/02 | 国际收支：Current account 的组成与符号 | Current-account components and signs

Current account 包括 trade in goods、trade in services、primary income 和 secondary income。出口记录为 credit，进口为 debit。Surplus 表示流入大于流出，deficit 相反。它是流量，不是政府“欠款总额”，也不等同于财政预算赤字。

知识点 02/02 | 国际收支：赤字原因和影响需结合背景 | Contextual causes and effects of a deficit

进口高可能来自高收入、强汇率或本国产品竞争力低；也可能是进口 capital goods，为未来出口能力打基础。赤字可能增加外债或使汇率承压，但居民可获得更多商品和机器。改善方法包括 depreciation、提高 productivity、贸易限制或降低 AD，各自有副作用。

Current account 的主体是本国居民与外国之间的流量，项目包括 goods、services、primary income 和 secondary income。Credit 带来收入，debit 造成支付。

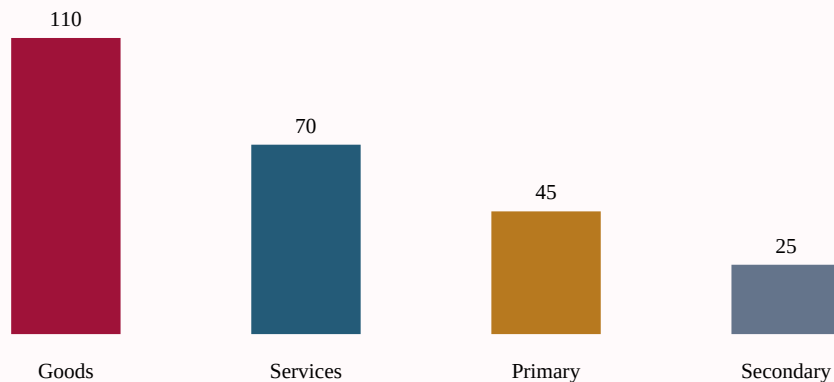
domestic income rises → consumers buy more imports → import expenditure rises → current-account balance may deteriorate; 若增长来自 export productivity，结果可能相反。

Goods exports 80, imports 100; services exports 40, imports 25; net primary income -5, transfers +2。Current balance = $(80 - 100) + (40 - 25) - 5 + 2 = -8$ ，为 deficit 8。

A current-account deficit means payments out exceed receipts in the recorded categories. It is not identical to government debt. A surplus may reflect strong exports, weak domestic demand or restricted imports, so its welfare effect is ambiguous.

Trade balance 只含 goods (有些口径含 goods and services 时必须看题目)；current account 范围更广。

Current account = goods + services + primary + secondary income



图解：经常账户把 goods、services、primary income 与 secondary income 的 credits 和 debits 相加。

2027-2029 考纲补充与整合 | Syllabus update and synthesis

补充知识点 01/02 | 跨国公司的东道国与母国影响 | MNCs in host and home countries

MNC 在多个国家拥有或控制生产。对 host country，它可能带来就业、资本、技术、出口和税收，也可能挤压本地企业、汇回利润、利用税制差异或造成环境成本。对 home country，它可能增加海外利润和全球竞争力，但若生产外迁，也可能减少某些本国岗位。结论取决于 linkages、技能转移、税收规则、监管与利润用途。

English explanation. A multinational company can transfer capital, technology and management skills to a host economy. However, the size of the gain depends on local sourcing, worker training, taxation and environmental regulation. For the home country, overseas production may lower costs and increase profits, but some domestic workers may face structural unemployment.

补充知识点 02/02 | 经常账户四部分 | Four components of the current account

current account balance = trade in goods + trade in services + primary income + secondary income. 出口与从国外收到的收入记作 credits；进口与付给国外的款项记作 debits。deficit 表示总 debits 大于 credits，不等于国家“没有钱”；surplus 也不自动表示生活水平更高。

English explanation. The current account records trade in goods, trade in services, primary income and secondary income. A depreciation may improve the balance by making exports cheaper and imports dearer, but the outcome depends on elasticities, contracts, imported inputs, productive capacity and time.

3. 核心词汇表 | Core vocabulary (共 41 词)

English term	中文准确口径	Exam-ready meaning
specialisation	专业化	Concentrating production on a narrower range of goods or tasks.
division of labour	劳动分工	Dividing production into separate specialised tasks.
absolute advantage	绝对优势	Producing more output with the same resources.
comparative advantage	比较优势	Producing at a lower opportunity cost.
opportunity cost	机会成本	The next best alternative forgone.

English term	中文准确口径	Exam-ready meaning
terms of trade	交换比率	The rate at which one product exchanges for another.
interdependence	相互依赖	Reliance of economies or firms on one another.
unit opportunity cost	单位机会成本	Output forgone to produce one extra unit of another product.
mutual gain	共同收益	A benefit received by both trading partners.
reciprocal	倒数	A number obtained by dividing one by another number.
free trade	自由贸易	International trade without artificial barriers.
protectionism	贸易保护主义	Government action restricting imports or supporting domestic producers.
tariff	关税	A tax placed on imports.
quota	配额	A legal limit on the quantity of imports.
embargo	禁运	A prohibition on trade with a country or in a product.
infant industry	幼稚产业	A new industry argued to need temporary protection.
retaliation	报复性措施	A trading partner's response to another country's restrictions.
globalisation	全球化	Increasing international integration of economies and markets.
multinational company (MNC)	跨国公司	A firm producing in more than one country.
outsourcing	外包	Purchasing activities from an outside supplier.
offshoring	离岸外移	Moving production or services to another country.
technology transfer	技术转移	The spread of production knowledge or equipment between economies.
repatriated profit	汇回利润	Profit transferred from a host country to the firm's home country.
labour standards	劳动标准	Rules and conditions governing workers' treatment.
exchange rate	汇率	The price of one currency in terms of another.
appreciation	升值	A rise in a currency's value under a floating system.
depreciation	贬值	A fall in a currency's value under a floating system.
export price	出口价格	The price foreign buyers pay for exported products.
import price	进口价格	The domestic-currency price of imported products.
demand for currency	货币需求	Demand created by purchases of a country's goods, services or assets.
supply of currency	货币供给	Currency offered to buy foreign goods, services or assets.
balance of payments	国际收支	A record of a country's transactions with the rest of the world.
current account	经常账户	The account recording trade in goods and services plus income and transfers.
visible exports	有形出口	Goods sold to other countries.
invisible exports	无形出口	Services sold to other countries.

English term	中文准确口径	Exam-ready meaning
current-account deficit	经常账户赤字	Current-account debits exceeding credits.
current-account surplus	经常账户盈余	Current-account credits exceeding debits.
competitiveness	竞争力	The ability to sell successfully against rival producers.
multinational company	跨国公司	A firm that owns or controls production in more than one country.
primary income	初次收入	Income flows such as profit, interest and wages between countries.
secondary income	二次收入	Current transfers between countries without a direct exchange of output.

4. 单元练习 | Unit practice (共 21 题)

练习 01/21

Question. A country is less productive than another in both goods. Can it still have comparative advantage?

练习 02/21

Question. Discuss whether greater specialisation benefits workers and consumers.

练习 03/21

Question. Country A gives up 2 rice for one phone; B gives up 5. Which exchange rate benefits both?

练习 04/21

Question. Using opportunity-cost data, explain why two countries may gain from trade.

练习 05/21

Question. Country A can produce 12 wheat or 6 cars; B can produce 8 wheat or 2 cars. Calculate the opportunity cost of one car in each country and identify comparative advantage. [5]

练习 06/21

Question. Which group is most likely to lose directly from a tariff on imported food?

练习 07/21

Question. Discuss whether an infant industry should receive protection.

练习 08/21

Question. Which is a likely host-country benefit of FDI: guaranteed equal income, technology transfer, no competition, or elimination of imports?

练习 09/21

Question. Discuss whether MNCs improve living standards in developing economies.

练习 10/21

Question. Who is most likely to gain from an appreciation: an exporter paid in foreign currency, a domestic tourist travelling abroad, or a producer competing with cheaper imports?

练习 11/21

Question. Discuss whether depreciation will reduce a current-account deficit.

练习 12/21

Question. The exchange rate changes from £ 1=\$1.40 to £ 1=\$1.20. Identify the change and calculate the new dollar price of a £ 250 export. [3]

练习 13/21

Question. Which transaction is a credit item: a resident buys an imported car, a foreign tourist buys a hotel service domestically, or government pays foreign interest?

练习 14/21

Question. Discuss whether a current-account deficit is harmful.

练习 15/21

Question. Goods exports are 70, goods imports 95, services exports 40, services imports 25, net primary income -4 and net transfers +1. Calculate the current-account balance. [4]

练习 16/21

Question. Country X gives up 2 units of rice to produce 1 car. Country Y gives up 5 units of rice to produce 1 car. Which statement is correct?

A X has comparative advantage in cars B Y has comparative advantage in cars C X must import cars D no trade can benefit either country

练习 17/21

Question. A country records goods -\$18bn, services +\$7bn, primary income +\$3bn and secondary income -\$1bn. Calculate the current-account balance. [3]

练习 18/21

Question. Explain how a tariff may protect domestic firms. [4]

练习 19/21

Question. Analyse how a depreciation may affect a manufacturer that imports components and exports finished products. [6]

练习 20/21

Question. Explain two possible benefits of an MNC to a host country. [6]

练习 21/21

Question. Discuss whether a current-account deficit should always cause a government to introduce import restrictions. [8]

5. 练习答案与解析 | Answers and explanations (共 21 题)

答案 01/21

正确答案 | Correct answer. Yes. It can have a lower opportunity cost in one good even without an absolute advantage.

答案 02/21

参考答案 | Model answer. Productivity and scale may lower prices and raise wages or choice. Workers gain expertise. However, repetitive work can reduce motivation, and demand changes may cause severe unemployment in specialised regions. Consumers become more dependent on supply chains.

答案 03/21

正确答案 | Correct answer. One phone for any amount strictly between 2 and 5 rice, such as 3 rice.

答案 04/21

参考答案 | Model answer. The country with the lower opportunity cost should specialise relatively more in that product. Combined output can rise because resources are used where sacrifice is smaller. At terms of trade between the two costs, the exporter receives more than domestically and the importer gives up less.

答案 05/21

正确答案 | Correct answer. A: one car costs $12/6 = 2$ wheat. B: one car costs $8/2 = 4$ wheat. A has comparative advantage in cars because its opportunity cost is lower; B has comparative advantage in wheat because one wheat costs $2/8 = 0.25$ car, compared with $6/12 = 0.5$ car in A.

答案 06/21

正确答案 | Correct answer. Domestic consumers, because they face higher prices and less choice, although local producers may gain.

答案 07/21

参考答案 | Model answer. Temporary protection may allow learning, scale economies and future competitiveness. It can preserve jobs and diversify the economy. But firms may become dependent and inefficient; consumers pay more and governments may be lobbied to extend protection. Conditions and a credible time limit matter.

答案 08/21

正确答案 | Correct answer. Technology transfer, although its extent depends on training and links with domestic firms.

答案 09/21

参考答案 | Model answer. They may create employment, wages, training, exports and tax revenue. Competition and technology can raise productivity. Conversely, profits may leave, local firms may close, jobs may be insecure and pollution may rise. Regulation and job quality determine the net effect.

答案 10/21

正确答案 | Correct answer. The tourist travelling abroad, because each unit of domestic currency buys more foreign currency.

答案 11/21

参考答案 | Model answer. Cheaper exports and dearer imports may raise export receipts and reduce import spending. Yet demand may be inelastic, firms may lack capacity, and imported raw materials raise costs. Short-run contracts can delay adjustment. Success depends on elasticities and time.

答案 12/21

正确答案 | Correct answer. The pound depreciates because each pound buys fewer dollars. The export price changes from \$350 to \$300. Whether export revenue rises depends on the response of foreign quantity demanded.

答案 13/21

正确答案 | Correct answer. The foreign tourist's purchase, because it is an export of services and brings foreign currency receipts.

答案 14/21

参考答案 | Model answer. It may weaken the currency, increase borrowing and indicate low competitiveness. Yet it may finance imported capital goods, reflect strong growth and provide consumers with choice. Harm depends on size, persistence, financing and whether imports improve future productivity.

答案 15/21

正确答案 | Correct answer. $(70 - 95) + (40 - 25) - 4 + 1 = -13$. The economy has a current-account deficit of 13 units; the negative sign is a flow deficit, not government debt.

答案 16/21

正确答案 | Correct answer. A – X has the lower opportunity cost of a car: 2 units of rice compared with 5 in Y. Comparative advantage depends on relative opportunity cost, not total output.

答案 17/21

正确答案 | Correct answer. Current-account balance = $-18 + 7 + 3 - 1 = -\$9\text{bn}$. The country has a current-account deficit of \$9bn.

答案 18/21

参考答案 | Model answer. A tariff raises the domestic price of imports. Consumers may switch towards domestically produced substitutes, increasing demand and sales for local firms. Domestic output and employment may rise, although consumers pay more and protected firms may face weaker pressure to become efficient.

答案 19/21

参考答案 | Model answer. Depreciation makes exported finished products cheaper in foreign currency, which may raise export demand and revenue. However, imported components become more expensive in domestic currency, raising production cost and possibly price. The net effect on profit depends on export and import shares, elasticities, contracts and the firm's ability to switch suppliers.

答案 20/21

参考答案 | Model answer. An MNC may create jobs, raising household income and tax revenue; if it trains workers, productivity and future employability also rise. It may bring capital and technology, expand productive capacity and exports, and create orders for local suppliers. Each benefit depends on local linkages and whether profits and skills remain in the host economy.

答案 21/21

参考答案 | Model answer. Restrictions may reduce imports, support domestic employment and improve the balance in the short run. However, tariffs and quotas raise consumer and producer costs, invite retaliation and reduce competition. A deficit may reflect productive investment or strong growth rather than weakness. Supply-side policies that raise productivity and export quality may be more sustainable. Intervention is more justified when a persistent deficit reflects low competitiveness and when measures comply with trade obligations.

6. 自学检查清单 | Self-study checklist (共 11 项)

- ☐ 能用机会成本而不是总产量判断 comparative advantage
- ☐ 能从产量表计算每单位产品的机会成本
- ☐ 能解释专业化和自由贸易的收益与风险
- ☐ 能区分 tariff、quota、subsidy 与 embargo
- ☐ 能评价 infant industry、employment、dumping、strategic industry 等保护理由
- ☐ 能分析全球化对消费者、工人、企业和政府的不同影响
- ☐ 能分别评价 MNC 对 host country 与 home country 的影响
- ☐ 能解释 fixed/floating exchange rate、appreciation 与 depreciation
- ☐ 能写出汇率变化影响出口进口的价格-需求链
- ☐ 能识别 current account 四部分并计算余额
- ☐ 能评价改善 current-account deficit 的政策与限制

附录 | Appendix

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